

Nashik District Heat Wave Action Plan 2026



DISTRICT ADMINISTRATION, NASHIK

**DISTRICT DISASTER MANAGEMENT
AUTHORITY, NASHIK**

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Figure 1: Trends in global average surface temperature between 1993 and 2022 in degrees Fahrenheit per decade. Most of the planet is warming (yellow, orange, red). Only a few locations, most of them in Southern Hemisphere oceans, cooled over this time-period.

(Source: NOAA)

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Executive Summary

The Nashik District Heatwave Action Plan has been formulated to address the growing challenge of extreme heat events in the district. The plan outlines a comprehensive framework for preparedness, mitigation, and response to reduce the adverse impacts of heatwaves on human health, infrastructure, and livelihoods. Developed in accordance with the guidelines of the National Disaster Management Authority (NDMA) and the provisions of the Disaster Management Act, 2005, the plan serves as a guiding document for various stakeholders to coordinate and effectively respond to heatwave situations. The key highlights of the plan, based on its chapter-wise content, are summarized below:

Chapter 1 is the introductory chapter that establishes the context of climate change and its contribution to the increasing frequency and intensity of heatwaves across the globe. It outlines the global and national trends of heatwaves and their devastating impact on human health, agriculture, and infrastructure. Additionally, it highlights the increasing vulnerability of India, particularly Maharashtra, to heatwaves and emphasizes the need for a structured action plan for districts like Nashik to mitigate the associated risks.

Chapter 2 sets the foundation for the District Heatwave Action Plan by emphasizing its rationale, objectives, and goals. It explains the importance of developing a heatwave action plan to minimize fatalities, heat-related illnesses, and infrastructure damage. The chapter also outlines the primary objectives of the plan, such as issuing early warnings, reducing exposure to heat, and ensuring the availability of healthcare facilities. The chapter further highlights the need for collaboration between various government departments, local bodies, and community organizations for effective implementation of the plan.

Chapter 3 provides a detailed profile of Nashik district, including its geographical location, climatic conditions, and population demographics. It highlights Nashik's increasing vulnerability to heatwaves, particularly in urban areas with dense population and limited green cover. The chapter also maps the high-risk areas and identifies vulnerable populations such as outdoor workers, elderly individuals, children, and people with pre-existing health conditions.

Chapter 4 elaborates on the early warning and heatwave alert system, primarily issued by the India Meteorological Department (IMD). It details the temperature thresholds for declaring

heatwaves and severe heatwaves in Nashik and introduces the color-coded heatwave warning system (Green, Yellow, Orange, and Red) used for different levels of heat risk.

Chapter 5 outlines a comprehensive set of preventive and mitigation measures to reduce the impact of heatwaves. It emphasizes public health measures such as establishing cooling centers, providing clean drinking water, ensuring adequate medical supplies, and conducting awareness campaigns. Additionally, the chapter recommends adaptive infrastructure measures such as promoting cool roofs, enhancing green cover, and improving ventilation in buildings. The importance of ensuring power supply, water availability, and providing medical assistance during heatwave conditions is also emphasized.

Chapter 6 focuses on the immediate actions to be taken when a heatwave alert is issued. It highlights the roles and responsibilities of various line departments such as health, water supply, electricity, transport, and public works to ensure a coordinated and rapid response. Measures such as deploying emergency medical teams, ensuring uninterrupted power supply, setting up temporary cooling centers, and providing heat-relief materials (ORS, water, shade) are outlined to protect the population during extreme heat conditions.

Chapter 7 emphasizes capacity building and training programs for various stakeholders, including healthcare professionals, community health workers, and frontline responders. It also outlines the importance of conducting public awareness campaigns to educate people on heatwave safety measures, recognizing symptoms of heat-related illnesses, and accessing emergency services. The plan encourages collaboration with non-governmental organizations (NGOs) and community groups to enhance outreach and impact.

Chapter 8, a critical component of the action plan, outlines the roles and responsibilities of various line departments. It ensures that all departments work in coordination during heatwave situations to minimize health risks, ensure water availability, maintain power supply, and provide emergency medical services. This chapter also mandates regular coordination meetings and review of the response actions taken during heatwaves.

Chapter 9, the final chapter provides a framework for monitoring, evaluation, and reporting (MER) to assess the effectiveness of the heatwave action plan. It outlines mechanisms for collecting real-time data on heat-related illnesses, deaths, and infrastructure disruptions.

Chapter 1: Background and status

Introduction

Climate change refers to long-term shifts in global weather patterns and temperature due to human activities, primarily the emission of greenhouse gases into the atmosphere. The climate crisis is not a distant concern for humanitarians. It is already unfolding before our eyes. In the past decade alone, more than 4,10,000 lives have been claimed by extreme weather events and climate-related disasters. Sadly, most of these victims resided in countries with low or lower-middle incomes.

RECENT TEMPERATURE TRENDS (1993-2022)

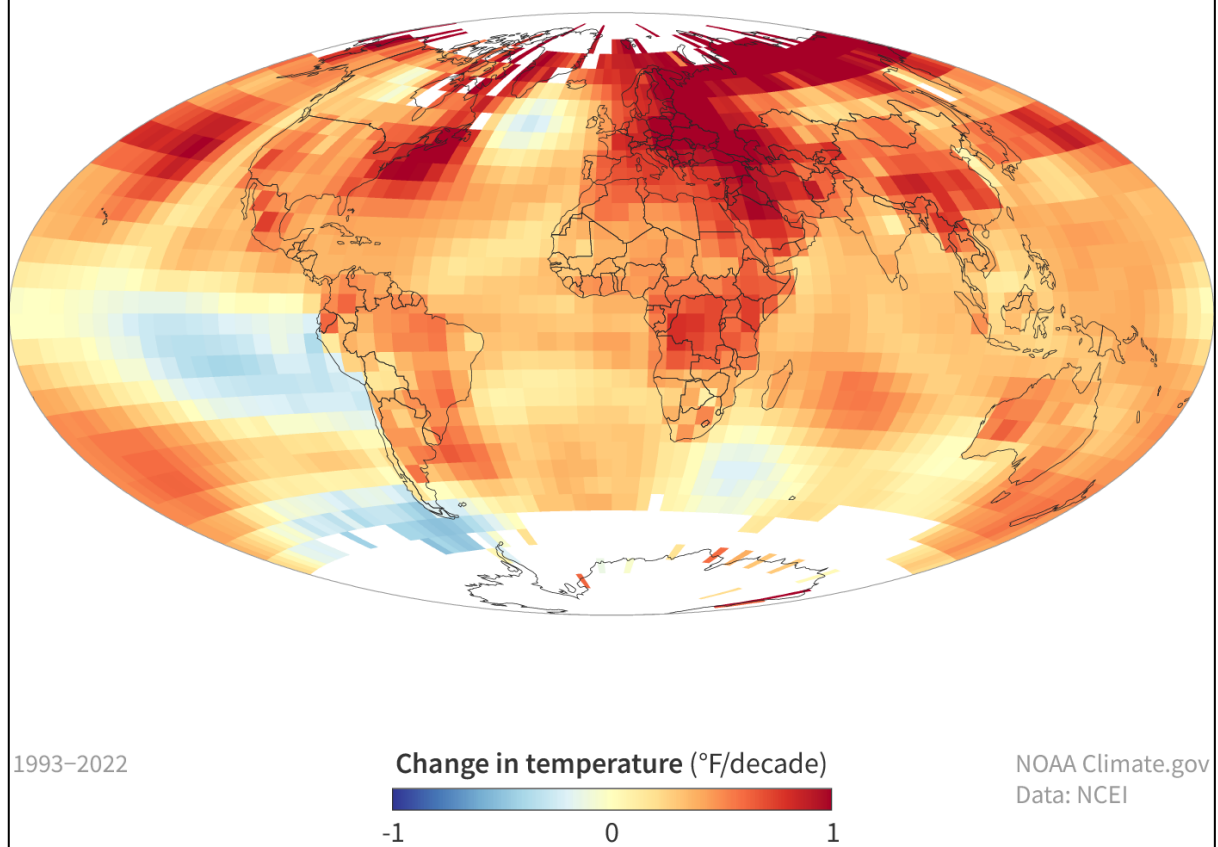


Figure 1: Trends in global average surface temperature between 1993 and 2022 in degrees Fahrenheit per decade. Most of the planet is warming (yellow, orange, red). Only a few locations, most of them in Southern Hemisphere oceans, cooled over this time-period. (Source: NOAA)

While humanitarian efforts often prioritize addressing the severe storms, droughts, and floods associated with a warming planet, there is another peril that receives less attention but inflicts

significant suffering: heat waves. Although they may not capture our imagination like major earthquakes or storms, heat waves have proven to be some of the deadliest disasters in recent times.

A heat wave refers to an extended period of excessively hot weather, often accompanied by high levels of humidity. It is characterized by a significant and prolonged increase in temperatures, well above the average for a particular region or season. Heat waves can vary in intensity and duration, ranging from a few days to several weeks. Exposure to extreme heat poses far greater dangers than is commonly understood.

The Russian heatwave of 2010, considered to be the most severe heat-related event on record by many scientists, resulted in a staggering death toll of up to 55,000 lives due to the combination of extreme heat and wildfires. Similarly, the 2003 heatwave in continental Europe is estimated to have caused over 70,000 deaths. South Asia has experienced the devastating impact of heatwaves in recent years, with reported deaths exceeding 3,800 during an eight-day heatwave in Bangladesh in 2008, and 2,000 deaths in India in 2015.

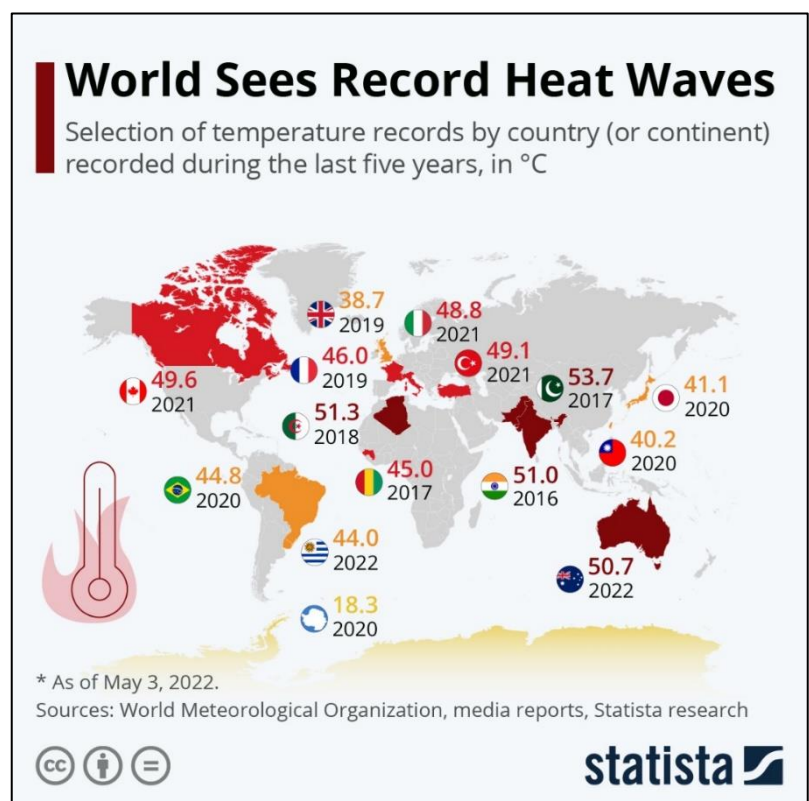


Figure 2: Temperature records recorded from 2017-2022
(Source: Statista)

Heatwave in India

The Intergovernmental Panel on Climate Change (IPCC) is an authoritative body that assesses scientific research on climate change and provides guidance to policymakers. One of the key findings of the IPCC report is the projection of increasing temperatures in India. Heatwaves are expected to become more frequent, longer, and more intense, posing risks to public health, agriculture, and livelihoods.

India experiences frequent and severe heat waves that can have significant impacts on human health, agriculture, infrastructure, and overall well-being. Heatwaves in India are a result of several factors, including geographical location, topography, and the influence of the monsoon season. During heatwaves, temperatures in India can soar to dangerous levels, often exceeding 40 degrees Celsius (104 degrees Fahrenheit) in many regions. The combination of high temperatures, intense solar radiation, and low humidity levels can create a hazardous environment, particularly for vulnerable populations such as the elderly, children, and individuals with pre-existing health conditions.

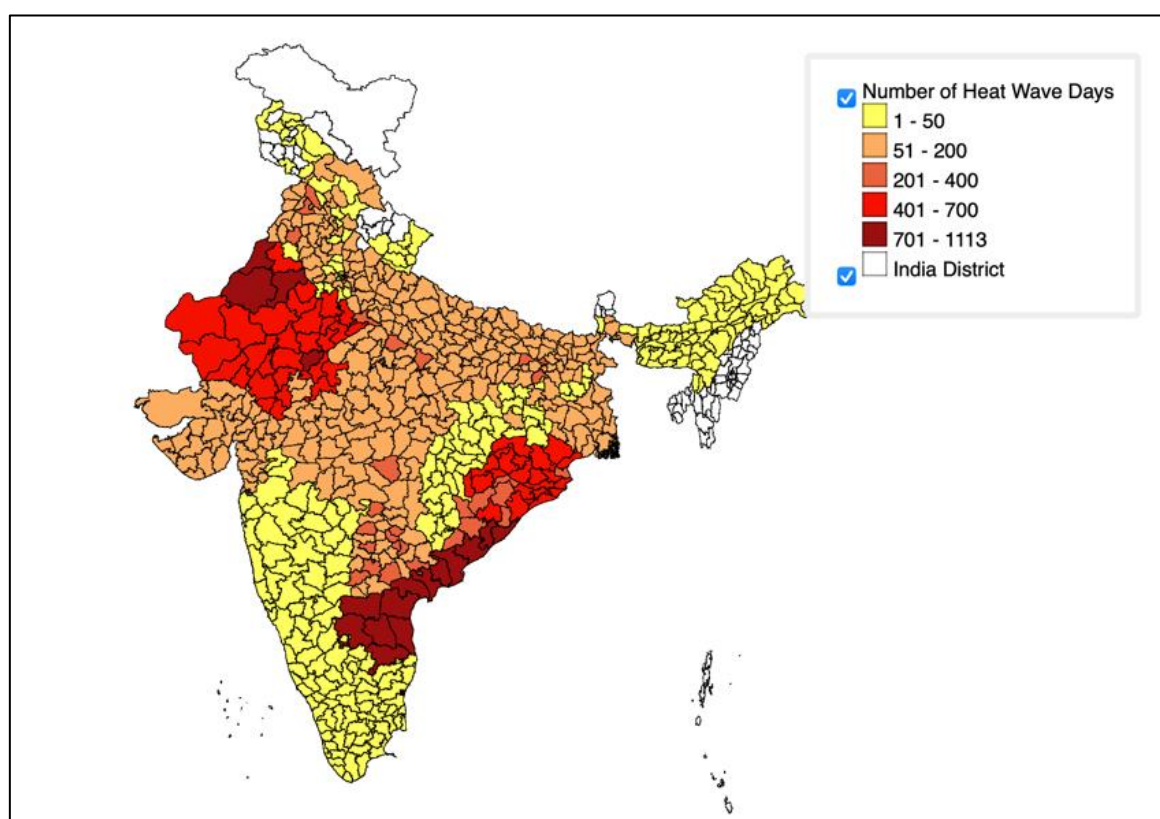


Figure 3: Total no. of Disastrous Heatwave Days from 1969-2019 (Source: IMD)

Over a span of 23 years, from 1992 to 2015, a staggering number of 24,223 lives were lost in India. However, it is crucial to acknowledge that this figure is likely a significant underestimate. Heatwave-related deaths are frequently underreported, primarily due to the limited number of countries maintaining comprehensive records of heatwaves and their resulting excess mortality.

Table 1: Year-wise details of recorded deaths caused by heat waves in India (Source: Beating the Heat by NDMA)

Year	Recorded deaths caused by Heatwave	Year	Recorded deaths caused by Heatwave	Year	Recorded deaths caused by Heatwave
1992	612	2000	534	2008	616

1993	631	2001	505	2009	1071
1994	773	2002	720	2010	1274
1995	1677	2003	807	2011	793
1996	434	2004	756	2012	1247
1997	393	2005	1075	2013	1216
1998	3058	2006	754	2014	1677
1999	628	2007	932	2015	2040

During episodes of intense heat, hospital admissions and fatalities linked to chronic conditions, such as diabetes and heart disease, inevitably surge. Unfortunately, these immediate causes often overshadow the underlying role of heat, leading to a failure to accurately document its critical impact on hospitalizations and deaths. Consequently, the true extent of the devastating consequences inflicted by extreme heat tends to remain concealed behind this reporting bias.

To address the impacts of heat waves, India has implemented various measures. These include heatwave early warning systems, public awareness campaigns, the establishment of cooling centres, and guidelines for outdoor workers to ensure their safety. Efforts are also being made to improve urban planning and infrastructure to mitigate the urban heat island effect and provide accessible cooling options. The states have provided guidelines to formulate their Heat Wave Action Plan.

Given the projected increase in frequency and intensity of heat waves due to climate change, it is essential for India to continue developing adaptation and mitigation strategies. These efforts involve implementing sustainable urban planning, promoting energy efficiency, and reducing greenhouse gas emissions to minimize the impacts of heatwaves on human health, agriculture, and overall resilience.

Impacts of Heatwave

The impact of heat waves extends to various facets, affecting human well-being, air quality, energy usage, agricultural productivity, water conservation, and the exacerbation of drought conditions. Exposure of crops to temperatures surpassing a critical threshold poses a substantial risk of crop failure. Moreover, heat waves contribute to elevated temperatures in urban

environments, creating urban heat islands, which, in turn, disrupt critical infrastructure networks. These occurrences not only influence the economy by diminishing labor productivity but also amplify the consequences of other climate-related hazards, such as droughts and forest fires.

Research underscores the multifaceted impacts of heat waves, encompassing health, infrastructure functionality, energy requirements, architectural considerations, water quality, and associated costs (Zuo et al., 2015). The interplay between these impacts further complicates the overall scenario, with factors like costs and energy consumption exhibiting interconnected dynamics. An investigation conducted by Azhar et al. (2017) delved into the vulnerability of India to heat waves. The findings revealed that, among the 640 districts assessed, 10 fell within the very high-risk category, while 97 were classified as high-risk. Notably, districts with heightened vulnerability were predominantly situated in the central regions of the country, characterized by lower urbanization levels, limited literacy rates, restricted access to water and sanitation, and a lack of household amenities. Some of the impact of heatwave are as follows:

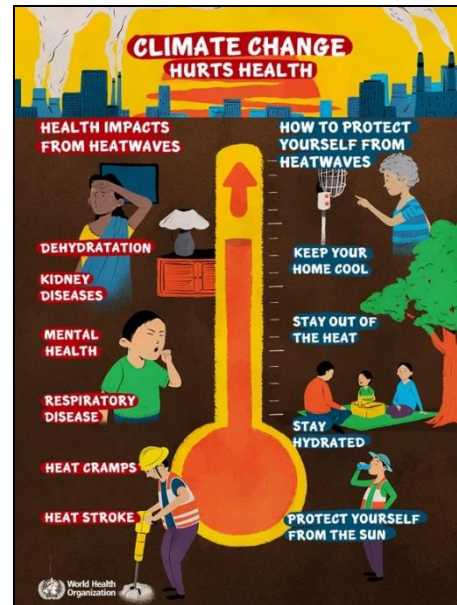


Figure 4: Effects and Prevention of Heat Wave. (Source: World Health

1. **Health Effects:** When the human body's temperature surpasses the optimal 37°C , the blood thickens, forcing the heart to work harder and potentially causing severe damage to vital organs. If the body's cooling mechanisms fail and the temperature continues to rise, dehydration and other seemingly mild symptoms of heat exhaustion set in. Without timely treatment, these symptoms can rapidly escalate into life-threatening heat stroke. It also causes disruption of the normal functioning of an ecosystem and impacts various sectors of society such as agriculture, energy, and transportation.
2. **Agricultural and Ecological Consequences:** Heatwaves severely threatens agriculture and ecosystems. During these events, extreme temperatures and drought conditions can cause crop failures, reduce yields, and impact livestock and fisheries. Heat stress can disrupt the growth, productivity, and nutritional quality of crops, affecting food security.

Additionally, heat waves can lead to forest fires, exacerbate desertification, and contribute to the loss of biodiversity and habitat degradation.

3. **Economic Implications:** The economic impact of heat waves is significant. Industries such as agriculture, tourism, and outdoor recreation experience substantial losses due to reduced productivity and visitor numbers. Heat-related health issues also strain healthcare systems, increasing medical costs. Furthermore, the strain on power grids during heatwaves can lead to higher energy consumption, increased electricity bills, and infrastructure damage, impacting businesses and households alike.
4. **Social Disruptions:** Heatwaves can disrupt daily life and social dynamics. High temperatures can limit outdoor activities, affect work productivity, and disrupt transportation systems. In urban areas, the urban heat island effect, where cities become significantly hotter than surrounding rural areas, intensifies the impact. Heatwaves can disproportionately affect marginalized communities with limited access to cooling facilities, amplifying existing social inequalities.
5. **Implication with Energy sector:** Heat waves exert significant impacts on the energy sector, with repercussions ranging from the compromised reliability of the energy grid to elevated costs and the potential for power outages. The escalation in electricity demand during heat waves, fuelled by increased usage of cooling devices like air conditioning and fans, places strain on the energy grid. This heightened demand poses a tangible risk of blackouts or brownouts, particularly if the energy supply cannot match the soaring demand. Furthermore, the efficiency of power plants, especially those reliant on water for cooling, is hampered by high temperature.

Chapter 2: Preparing Heatwave Action Plan

Heatwave and Disaster Management

Under section 2(d) of the Disaster Management Act, 2005, the term ‘Disaster’ is defined as a catastrophe, mishap, calamity or grave occurrence in any area, arising from man-made or natural causes, leading to substantial loss of life or human misery, or damage to, or destruction of property or environment, and as such, the impacts are beyond the coping capacity of the local community and requires external support. Heatwave is not a notified disaster by the Government of India, that is, it is not among the 12 disasters which are eligible for relief under the National/ State Disaster Response Fund norms. However, the state government can utilize 10 percent of the funds available to them under SDRF for providing immediate relief to the affected by notifying it as a state specific disaster.

Rationale Behind the Plan

The district of Nashik, located in the state of Maharashtra, has been experiencing an increase in the frequency, intensity, and duration of heatwaves in recent years. This rise in extreme heat events poses significant challenges to public health, livelihoods, infrastructure, agriculture, and overall socio-economic stability in the district. Recognizing the adverse impacts of heatwaves, it becomes imperative to develop a structured and comprehensive District Heatwave Action Plan (DHAP) to enhance preparedness, response, and mitigation measures.

The primary rationale behind formulating this action plan is to minimize the loss of life, prevent heat-related illnesses, and reduce the socio-economic impact of heatwaves on vulnerable populations, including outdoor workers, children, elderly, persons with disabilities, and those with pre-existing health conditions. The plan will also aim to safeguard critical infrastructure, ensure the continuity of essential services, and strengthen community resilience to heatwaves.

The formulation of the District Heatwave Action Plan aligns with the National Action Plan on Climate Change (NAPCC), National Disaster Management Authority (NDMA) Guidelines for Preparation of Action Plans for Prevention and Management of Heatwaves, 2016, and the Disaster Management Act, 2005. Additionally, it seeks to incorporate local learnings, community experiences, and the district administration’s response capacity to tailor the strategies to Nashik’s specific geographical and socio-economic context.

The District Heatwave Action Plan will serve as a guiding document to:

1. Establish early warning systems and dissemination channels to alert communities during extreme heat events.
2. Strengthen healthcare infrastructure to manage heat-related illnesses and reduce mortality rates.
3. Develop community-centric strategies to reduce exposure and vulnerability to heat stress, especially for high-risk groups.
4. Build coordination mechanisms among various government departments, local bodies, and non-governmental organizations for effective preparedness and response.
5. Promote adaptive measures such as the development of cool-roof strategies, establishment of heat shelters, provision of drinking water, and improving public spaces to mitigate the heatwave impact.

Objective of the plan

The plan's main goal is to mitigate the impacts of heat-related illnesses and fatalities by issuing health alerts, with a special focus on the most susceptible populations. It involves timely dissemination of advice and notifications regarding upcoming heat-wave conditions, creating awareness among the general public and healthcare professionals to take necessary precautions. Additionally, the plan aims to efficiently coordinate and mobilize all accessible resources to proactively prevent and minimize the adverse health effects of heat waves. To achieve this objective, the plan provides a structured framework for implementation, coordination of a comprehensive response, and ongoing evaluation of activities aimed at addressing extreme heat situations.

Goals

- Ensure effective public health promotion, early warning systems, and informative campaigns to alert individuals about potential hazards and educate them on preventive measures to safeguard themselves and reduce the risk of heat-related health issues.
- Provide comprehensive training to caregivers in social institutions like nursing homes and orphanages on protective measures, recognizing relevant symptoms, and managing cases related to heat-related conditions.

- Implement sustainable methods to supply clean water to high-risk areas while minimizing the use of single-use plastics.
- Distribute essential items such as ice packs, oral rehydration salts, and heat protection kits containing items like umbrellas, towels, and fans.
- Establish proactive measures within the healthcare system, including stockpiling necessary medications and intravenous fluids, and establishing temporary outreach clinics in vulnerable regions.
- Designate shaded areas and cooling options, such as misting systems at gathering points or utilizing religious sites and public spaces as cooling centres. Implement strategies like painting roofs white or adopting "cool roofing" interventions and enhancing passive ventilation in homes.
- Issue policy guidelines to suspend work, provide free public transportation, or adjust working hours to reduce heat exposure.
- Implement measures to ensure uninterrupted power supply to critical facilities like water purification systems.
- Direct targeted cash distributions or suspend utility and other payments to alleviate financial burdens.
- Implement measures to protect crops and animals, which are vital resources for affected communities.
- Prioritize care for the most vulnerable populations, including visits from community healthcare workers to pregnant women or elderly individuals living alone.

Chapter 3: Heatwave in Nashik

Nashik Profile

About the District

Nashik District is located between 18.33 degrees and 20.53 degrees North latitude and between 73.16 degree and 75.16 degree East longitude in the Northwestern part of the Maharashtra. Nashik is at 565 meters above mean sea level. The district has great mythological background. Lord Rama lived in Panchvati during his vanvas. Agasti Rushi also stayed in Nashik for Tapasya. The Godavari River originates from Trimbakeshwar in



Figure 5: The political map of Nashik District

Nashik. One of the 12 Jyotirlingas also at Trimbakeshwar. Nashik has to its credit many well-known and towering personalities like Veer Sawarkar, Anant Kanhere, Rev. Tilak, Dadasaheb Potnis, Babubhai Rathi, V.V. Shirwadkar and Vasant Kanetkar and more. Nashik is also known as Mini Maharashtra, because of the climate and soil conditions of Surgana, Peth, Igatpuri resemble Konkan. Niphad, Sinnar, Dindori, Baglan blocks are like Western Maharashtra and Yeola, Nandgaon, Chandwad blocks are like Vidarbha Region. Nashik, Malegaon, Manmad, Igatpuri are some of the big cities situated in the Nashik District.

Nashik district is the third largest district in Maharashtra in terms of Population of 61,09,052 and area occupying an area of 15,582 square kilometers in the north Maharashtra region. It is bounded by Dhule district to the north, Jalgaon district to the east, Aurangabad district to the southeast, Ahmednagar district to the south, Thane district to the southwest, Valsad and Navsari districts of Gujarat to the west, and Dang district to the northwest. The Western Ghats or Sahyadri range stretches from north to south across the western portion of the district. Except for the few westernmost villages, the western portion is hilly, and intersected by ravines, and

only the simplest kind of cultivation is possible. The western slope of the Ghats is drained by several rivers, including the Daman Ganga River, which drains westwards to the Arabian Sea.

The larger eastern portion of the district, which lies on the Deccan Plateau, is open, fertile, and well cultivated. The Satmala-Chandwad Range, which runs east and west, forms the chief divide of the plateau region. Peninsular India's largest river Godavari originates in the district in the

Trimbakeshwar Range and continues its sojourn eastwards through the district. The Satmala-Chandwad Range forms a watershed, such that, the rivers emerging to its south drain into the Godavari. These include the Kadva and Darna both of which are tributaries of the Godavari. To the north of the Satmala-Chandwad Range, the Girna River and its tributary, the Mosam, flow eastward through fertile valleys into the Tapti River. The Trimbakeshwar Shiva Temple is located in Trimbak, one of the twelve Jyotirlingas, where the Hindu genealogy registers at Trimbakeshwar, Maharashtra are kept. The origin of the sacred Godavari river is near Trimbak.

Vulnerability of Nashik to Heatwave

Nashik experiences high temperatures during the summer months, with the potential for heatwaves that can adversely affect health, agriculture, and water resources. While Nashik generally has a moderate climate, there have been instances of temperatures rising significantly, leading to heat-related illnesses. The North-Eastern blocks of Malegaon and Nandgaon are particularly vulnerable to the Heat wave and come under the red category. The Baglan, Yeola, Chandwad, Deola, Sinnar, Niphad and Kalwan blocks are moderately prone to Heat wave. The map below shows the vulnerable areas of District Nashik to Heat Wave.

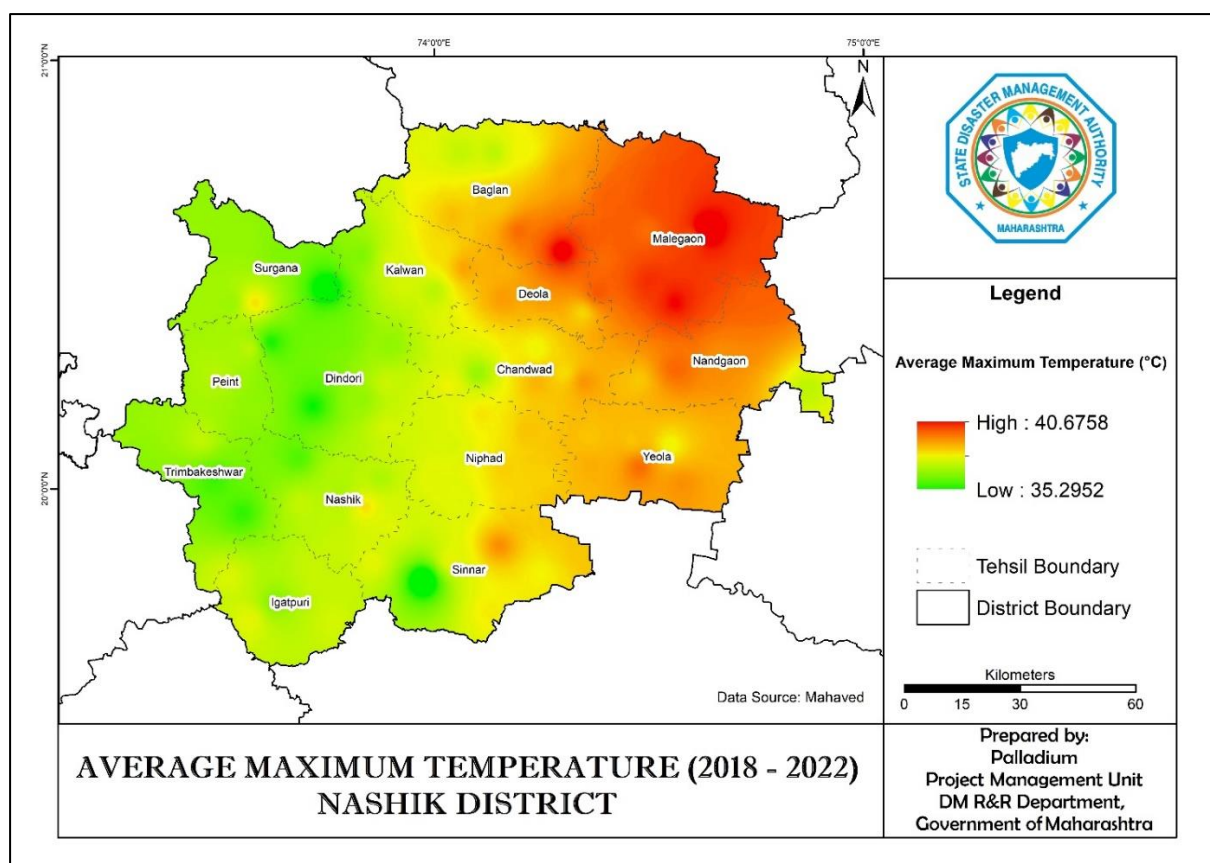


Figure 7: Heatwave vulnerability map of Nashik

For Heatwave, the government of Maharashtra has set instructions for the district administration and has defined basic roles and responsibilities of different line departments as mentioned below:

General Instructions for the District Authority

1. Heat wave management period will generally be from 1st March to 15th June every year.
2. Contact numbers 100, 101, 102, 104, 108, 112, 1077, and 1070 should be circulated to citizens for awareness of emergency numbers.
3. The Collector should convene a meeting of the District Disaster Management Authority before March 15 and prepare an action plan for the heat wave period (March 1 to June 15). While preparing the said action plan, determine the heatstroke prone areas by taking into consideration the record of day and night temperature of the last five years, humidity and dust particles (SPM) in the area, slums, crowded places, workers working in factories, businesses, beet furnaces and similar businesses, places of pilgrimage in rural areas, religious places, etc.

4. For health-related matters, the District Surgeon of the Dhule district will act as the Liaison Officer (Nodal Officer) and they will be in constant contact with the District Health Officer to ensure that all the health facilities in every Sub-Centre, Primary Health Center (PHC), and Civil Hospital are available full-time. The Collector will review the same on every Monday of the week and prepare the information in the attached form (Form 1).
5. Early Warning is forecasted through India Meteorological Department (IMD). In order to collect the latest information related to this, the District Regional Officer or the District Pollution Control Board Officer of each district will take the information from the Indian Meteorological Department (IMD) and convey it to the Collector every day. As soon as the heat wave warning is received from the Meteorological Department, the Collector's office will share the warning with the public via a press note, local television channels, radio, WhatsApp or Alert System (CAP), etc.
6. During the heatwave period, it is noticed that all patients with fever, body rash, nausea, vomiting, weakness, fatigue are treated with generic antibiotics. But in case of symptoms like above, treatment related to heat stroke is necessary. The district surgeon and district health officer, their subordinate health officials and staff (Medical and Para Medical Staff) must be trained in the plan and treatment related to heat stroke. This training should be conducted twice in the third week of March and in the second week of April to ensure that the patients get proper treatment. Sometimes a patient is admitted to the hospital in an unconscious state or a patient is brought who has died before admission or is declared dead after admission and the causes of death are recorded as heart attack, over stress, low or high blood pressure, febrile illness etc., even though death has actually occurred due to heat stroke. Therefore, accurate statistics are not available as the said death is reported to be due to other causes. This makes it difficult for the government to formulate a policy regarding heatwave, so if a patient shows the above symptoms during the heat wave warning period from 1st March to 15th June and the patient is not suffering from any other infectious disease or other chronic disease, it becomes important to ensure that the patient died due to heat stroke. Important factors to note are- Is the area where the patient lives heat prone, place of work, nature of work / occupation, age and other similar factors. The medical officer of the hospital where the death is registered will be able to declare the patient died of heat stroke.
7. From March 1, a separate register should be kept for the patients admitted due to heat stroke in all government hospitals and a report should be submitted to the District

Collector after making a detailed record of the patients admitted for treatment due to heat stroke and the persons who die due to heat stroke. The prescribed specimen of the register is given in Form 1.

8. Some beds should be reserved in government hospitals in urban areas as Cold Room/Wards. Information about the same should be announced as well as the same should be provided to the District Collector.
9. The concerned authorities should ensure that the fans are kept on and the air circulates in all the railway stations and bus stations.
10. In case of prior warning of heat wave, the power distribution company should ensure that there is no interruption of power supply / load shedding during afternoon hours.
11. Care should be taken to ensure that drinking water system, fans and first aid facilities are available in places like market committees, religious places, places of pilgrimage etc.
12. Care should be taken of the laborers working under the National Rural Employment Guarantee Scheme. Their work should be divided into shifts (Morning and Evening).
13. Care should be taken to ensure that all gardens/parks in urban areas are not closed during the afternoon. These places must be available for rest during the afternoon for citizens coming from rural areas to the city for work, roadside vendors and itinerant traders, daily wage workers.
14. Police personnel above 50 years of age, ST bus conductors and drivers and similar employees should be given convenient working hours keeping in mind the above matters.
15. If prior notice is given regarding heat wave, the above factors should be taken into consideration while allowing marches, demonstrations between 2 pm to 4 pm.
16. 16. The owners/traders/factories concerned should be instructed to give rest after certain period to the laborers who have to work in front of the furnace.
17. Fire audits to be conducted in hospitals, schools, factories, industries etc. and other places of importance.
18. Public awareness program among the citizens will be implemented by the local administration regarding timely and adequate provision of drinking water and shade to domestic animals.
19. The District Information Officer will be in constant contact with the District Disaster Management Officer and convey the advance notices and warnings related to the heat wave through various channels to alert the people of the concerned areas.

Roles and Responsibilities of Line Department

Table 2: Roles and Responsibilities of Line Departments

Department	Roles and Responsibilities
District Disaster Management Authority	• Conduct review meetings every 15 days between March to June. • To coordinate with the departments and check whether relevant actions have been implemented. • Awareness about emergency numbers - 1077, 112, 102, 104, 108 • To disseminate the Do's and Don't's related to heatwave. • Develop a policy-based "Heatwave Action Plan", Standard Operating Procedures (SoPs) and Incident Response System (IRS) which will detail the roles and responsibilities of various stakeholders to mitigate the impact of heatwave on the lives and livelihoods of the people. • The District Disaster Management Authority will depute Heatwave Response Officers at the district level in the event of heatwave. • Contact NGOs, CSOs and CSRs to voluntarily donate umbrellas, footwear and cotton towels for the larger population. • Use the existing Information Office to disseminate IEC materials through social media.
District Health Officer	• Conduct fire safety audits of health centers. • To provide training for para-medical staff on procedures to be followed in case of heat stroke. • Prepare a departmental heat wave action plan based on the national guidelines. • A nodal officer should be appointed to monitor and report heat wave related diseases. • Information boards regarding prevention of heat wave should be placed in the premises of every health center. • Separate medical wards should be arranged in health centers for heatstroke patients. • A Rapid Response Team (RRT) should be formed for the treatment of heat stroke patients. Multi-purpose workers, ASHA workers should be instructed to collect aid and details of heat stroke patients. Available information should be forwarded to the Taluka Health Officer. • Ambulances should be kept ready at public places, especially in the afternoon. • Primary Health Centers (PHCs) and other local hospitals should be adequately stocked with essential medical supplies including ORS powder etc.

Department	Roles and Responsibilities
	• Special care should be provided for vulnerable groups such as children, disabled persons, the elderly, the elderly and labourers. • Heatstroke-related cases and deaths should be reported daily. • Department's no objection certificate should be made mandatory for large events where food is served. • Identification of heat related illnesses or deaths and reporting them separately in the prescribed format.
Superintendent of Police	• Shelters or booths should be erected at signals for police. • First aid kits should be made available at police stations and traffic booths. • Heat stroke first aid and CPR training should be conducted for one policeman at every police station. • Do not issue permission letter for open events organized in the afternoon. • In any public event, provision of abundant water should be allowed as per the availability of space. • According to warnings from the Meteorological Department, large events should not be allowed on hot days.
Municipal Corporations and Councils	• Drinking water should be provided at public places like markets, main offices, bus stands, taxi stands, security stands etc. • Fans/coolers etc. should be repaired if they are not working properly • Posters regarding heat wave precautions should be posted at public spaces. • No parks should be closed between 12 pm to 4 pm.
Zila Parishad	• Spread awareness and measures to protect against heat waves at the Panchayat level. • Schedule working hours of MREGA workers and provide cold water at work places. • Organize training sessions at block level in coordination with the Health Department. • Use Gram Sabha as an effective medium for creating awareness about heat waves.
School Education	• According to the warnings, plan the opening of schools/colleges and provide facilities for cooling, first aid and drinking water. • Change the school opening hours and give holidays according to the heat wave situation. • Students should avoid field/physical activities in summer between 11pm- 4pm • Educate students to protect themselves from heat wave and provide contact details of emergency services.

Department	Roles and Responsibilities
	• Assembly should be conducted in the morning session only. • Ensure that the fans and coolers in schools are in working conditions. • Hydrating liquids such as Sherbat, buttermilk and ORS packets may be provided to the children under the mid-day meal scheme.
Labour	• Provide necessary facilities to reduce the impact of heat waves on workers. • Schedule the working hours of workers • Provide hot and cold water at the workplace. • Provide heat stroke first aid and CPR training to workers.
Transport	• Public transport services will remain smooth during peak heat wave. • Facilities like water, fans and toilets will be provided at waiting areas. • Precautionary measures will be displayed on the buses and information leaflets will be distributed to the passengers. • Public address system of bus shelters etc. will be used to broadcast emergency videos, IEC. • Ensure that first aid kits (with ample stock of ORS) are available in all buses during summer. • Water facilities, fans will be in good condition at bus shelters.
Forest	• There should be adequate tree plantation and water supply for wild animals and birds. • Monitor the areas to prevent fire and take necessary preventive measures.
Water Supply and Sanitation	• Provide drinking water facilities in rural areas and repair boreholes and ponds urgently. • Inspect and maintain all types of water sources before the summer season. • Plan for water supply in the morning and evening. These instructions should be given at the Gram Panchayat level, especially in the context of public water supply.
Public Works	• Promote the use of cool roof paint in government buildings and institutions. • Provide shelter and first aid kits for the injured at construction sites.
Power Supply	• All measures should be taken to ensure uninterrupted power supply during heat waves.

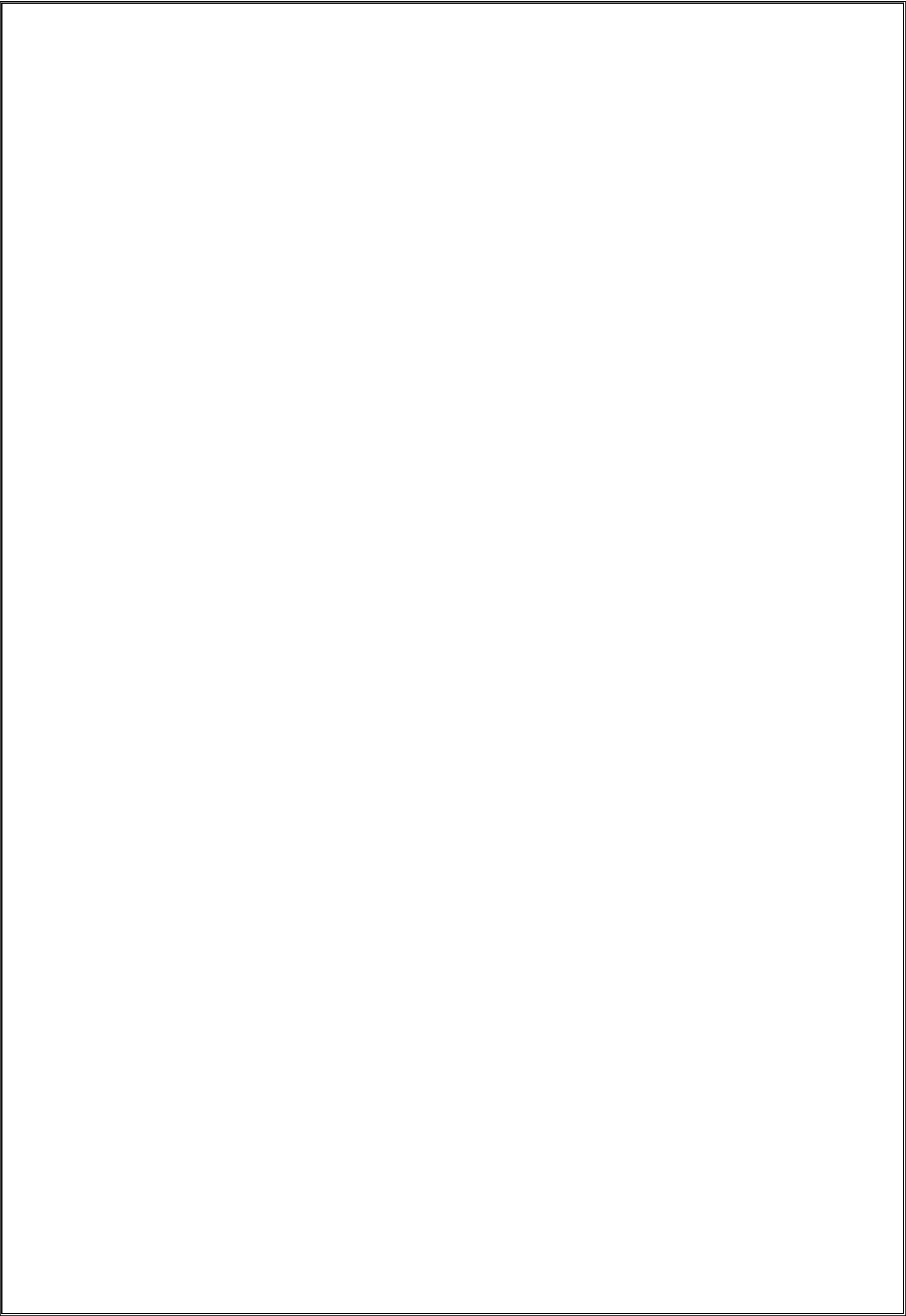
Department	Roles and Responsibilities
	IEC material related to heat waves should be disseminated on electricity bills and IEC-related information sheets should be distributed along with electricity bills.
Agriculture	- Spread awareness among farmers about heatwave and its precautions. - Promote the use of heat resistant crops if available.
Women and Child Development	- To conduct heat stroke first aid and CPR training for ASHA workers and Anganwadi workers. - Under ICDS scheme, Sarbat, Buttermilk and ORS packets will be provided to children and pregnant and lactating women. - To create awareness about heat wave at Gram Panchayat level through ASHA workers and Anganwadi workers.
Maharashtra Industrial Development Corporation (MIDC)	- To provide adequate shelter and cold drinking water facilities to workers in the industries. - To provide heat stroke first aid and CPR training to the security officers. Heat wave safety notices should be displayed outside every industry. - Include a chapter on heat wave preparedness in the emergency plan of the industry. - Industrial areas are prone to fire incidents, so adequate facilities like water storage and foam tenders should be maintained.
Animal Husbandry	- Prepare notice boards and information leaflets to inform the local people about the safety of animals during heat waves. - Organize awareness programmes for animal protection and maintain adequate stock of veterinary medicines and provide water and facilities for animals. - In areas with large numbers of livestock, external fodder and cold water will be made available.

Vulnerability Assessment

Heatwaves pose a significant threat to human health, particularly for certain segments of the population who are more vulnerable due to various factors. As global temperatures continue to rise, it becomes crucial to identify and prioritize the needs of these individuals during extreme heat events.

- Elderly Population:** The elderly, especially those above the age of 65, are among the most vulnerable during heatwaves. They often have decreased physiological resilience, chronic health conditions, and limited mobility, making it challenging for them to adapt to extreme heat. Additionally, many older individuals take medications that can affect their thermoregulation, making them more prone to heat-related illnesses.

2. **Infants and Young Children:** Babies and young children are more susceptible to heat stress due to their underdeveloped thermoregulatory systems and limited ability to communicate discomfort. Their bodies produce more heat and have a reduced capacity to cool down, putting them at a higher risk of dehydration, heat exhaustion, and heatstroke. Providing extra care and ensuring adequate hydration during heatwaves for this age group is essential.
3. **Individuals with Chronic Medical Conditions:** People with pre-existing chronic medical conditions, such as cardiovascular diseases, respiratory illnesses, diabetes, and obesity, face increased vulnerability during heatwaves. Extreme heat can worsen these conditions and lead to complications, as the body's ability to regulate internal temperature is compromised. Additionally, those relying on medical devices that require electricity may face additional challenges during power outages caused by heat-related events.
4. **Socioeconomically Disadvantaged Groups:** Socioeconomically disadvantaged populations, including the homeless, low-income households, and individuals living in substandard housing, are at higher risk during heatwaves. Limited access to air conditioning, insufficient resources for cooling measures, and the inability to afford alternative accommodation in cooler environments make them more vulnerable to heat-related health issues. These groups may also face challenges in accessing healthcare services and obtaining necessary medications during extreme heat events.
5. **Outdoor Workers:** Outdoor workers, such as construction workers, agricultural laborers, and first responders, face elevated risks during heatwaves due to prolonged exposure to high temperatures. Their occupation often requires physical exertion, which increases heat production and limits the body's ability to dissipate heat. Inadequate access to shade, breaks, and hydration resources can further exacerbate the risks associated with their work environments.
6. **Individuals with Mental Health Conditions:** People with mental health conditions may be more vulnerable to the effects of heatwaves due to factors such as medication side effects, limited self-care capacity, and challenges in recognizing and responding to heat-related symptoms. Heat stress can exacerbate mental health symptoms, increase agitation and irritability, and contribute to heightened anxiety levels.



Chapter 4: Heatwave alert and Warning System

To mitigate the adverse effects of extreme heat on human health and facilitate effective response measures, an early warning system for heatwaves has been developed in India. This system aims to provide timely information and alerts to the public, policymakers, and relevant stakeholders to help them prepare and take necessary precautions.

The early warning system for heatwaves in India consists of the following key components:

Meteorological Monitoring: Meteorological agencies, such as the India Meteorological Department (IMD), monitor temperature patterns across the country using a network of weather stations and satellite data. This enables them to identify regions where temperatures are expected to reach critical levels and issue heatwave alerts.

Heatwave Criteria and Thresholds: The IMD has established specific criteria and thresholds to define a heatwave in different regions of India. These criteria consider factors like maximum temperature, duration, and departure from normal temperatures. When the observed conditions meet or exceed these thresholds, a heatwave alert is issued.

Heatwave Early Warning: Once a potential heatwave is identified, the IMD issues early warnings to the respective state and local authorities, as well as media outlets. These warnings provide advance notice and relevant information, including the expected duration, intensity, and geographical areas likely to be affected by the heatwave.

Dissemination of Information: The IMD collaborates with various agencies, including disaster management authorities, health departments, and media channels, to disseminate heatwave alerts and advisories. This ensures that the information reaches a wide range of audiences, including vulnerable communities, policymakers, healthcare providers, and the public.

Public Awareness and Preparedness: Public awareness campaigns are conducted to educate individuals on the risks associated with heatwaves and the necessary preventive measures. These campaigns emphasize the importance of staying hydrated, seeking shade, wearing right clothing, and avoiding excessive outdoor activities during peak heat hours.

Heatwave Response Measures: The early warning system helps authorities and healthcare institutions to activate heatwave response plans promptly. This may include opening cooling centers, providing access to drinking water, establishing dedicated helplines, and mobilizing healthcare services in areas expected to be severely impacted by the heatwave.

Temperature/Humidity Index

The Temperature/Humidity Index (THI) is a widely used metric to assess heatwave conditions, considering both temperature and humidity levels. In India, where heatwaves are a common occurrence, the THI provides valuable insights into the severity of heat stress on human health.

- 1. Understanding the Temperature/Humidity Index (THI):** The THI combines temperature and relative humidity measurements to provide a comprehensive assessment of heat stress. It considers the body's ability to dissipate heat through sweat evaporation. Higher THI values indicate more severe heat stress, making it an essential tool for evaluating the potential impact of heat waves on human health. The Heat Index chart is given below:

Table 3: The Heat Index Chart

		Temperature (°F)															
Relative Humidity (%)		80	82	84	86	88	90	92	94	96	98	100	102	104	106	108	110
	40	80	81	83	85	88	91	94	97	101	105	109	114	119	124	130	136
	45	80	82	84	87	89	93	96	100	104	109	114	119	124	130	137	
	50	81	83	85	88	91	95	99	103	108	113	118	124	131	137		
	55	81	84	86	89	93	97	101	106	112	117	124	130	137			
	60	82	84	88	91	95	100	105	110	116	123	129	137				
	65	82	85	89	93	98	103	108	114	121	128	136					
	70	83	86	90	95	100	105	112	119	126	134						
	75	84	88	92	97	103	109	116	124	132							
	80	84	89	94	100	106	113	121	129								
	85	85	90	96	102	110	117	126	135								
	90	86	91	98	105	113	122	131									
	95	86	93	100	108	117	127										
	100	87	95	103	112	121	132										

(Source: National Oceanic and Atmospheric Administration)

- 2. Heatwave Classification:** In India, the THI is often utilized to classify heatwave events and issue appropriate warnings. Different regions in the country may have varying thresholds for defining a heatwave based on the THI, considering the local climate and acclimatization of the population. For instance, the India Meteorological Department (IMD) uses the THI to categorize heatwave conditions into different levels of severity, such

as moderate, severe, and extreme, aiding in effective heatwave management and public awareness.

- 3. Impact on Human Health:** The THI plays a crucial role in assessing the potential health risks associated with heat waves. Higher THI values indicate increased heat stress on the human body, leading to a higher likelihood of heat-related illnesses such as heat exhaustion and heatstroke. By monitoring the THI, health authorities can issue advisories and implement appropriate measures to protect vulnerable populations, including the elderly, children, and individuals with pre-existing health conditions.
- 4. Urban Heat Island Effect:** The THI is particularly relevant in urban areas where the urban heat island effect amplifies heatwave conditions. Urban environments, with concrete structures and limited green spaces, tend to retain more heat, resulting in higher THI values compared to rural areas. This effect poses additional challenges for heatwave management in densely populated cities, emphasizing the importance of monitoring THI levels and implementing heat resilience measures.

Criteria to declare Heatwave

The following criteria are used to declare a prevailing heat wave condition:

- i. Based on Departure from Normal
 - Heat Wave: Departure from normal is 4.5°C to 6.4°C.
 - Severe Heat Wave: Departure from normal is >6.4°C.
- ii. Based on Actual Maximum Temperature (For plains only)
 - Heat Wave: When actual maximum temperature = 45°C.
 - Severe Heat Wave: When actual maximum temperature = 47°C.
- iii. If the maximum temperature of a station reaches at least 40°C or more (for plains)
- iii. If the maximum temperature of a station reaches at least 37°C or more (for coastal areas)
- iv. If the maximum temperature of a station reaches at least 30°C or more (for hilly regions)

To declare a heat wave, one of the above criteria should be met for in at least two stations in a meteorological sub-division for at least two consecutive days. A heat wave will be declared on the second day.

Colour codes for Heatwave Alert:

IMD currently follows a single system of issuing warnings for the entire country. Heat waves are classified into different severity based on the temperatures. The table below shows the different colour codes, with their meaning, impact and actions needed.

Table 4: Different colour codes, with their meaning, impact and suggested actions.

Colour Code	Alert	Warning	Impact	Suggested Action
Green (No action)	Normal Day	Maximum temperatures are near normal	Comfortable temperature. No cautionary action required.	Normal activity
Yellow Alert (Be updated)	Heat Alert	Heat wave conditions at isolated pockets persists for 2 days	Moderate temperature. Heat is tolerable for general public but moderate health concern for vulnerable people e.g. infants, elderly, people with chronic diseases	(a) Avoid heat exposure. (b) Wear lightweight, light-coloured, loose, cotton clothes. (c) Cover your head
Orange Alert (Be prepared)	Severe Heat Alert for the day	(i) Severe heat wave conditions persists for 2 days (ii) Through not severe, but heat wave persists for 4 days or more	High temperature. Increased likelihood of heat illness symptoms in people who are either exposed to sun for a prolonged period or doing heavy work. High health concern for vulnerable people e.g. infants, elderly, people with chronic diseases.	(a) Avoid heat exposure- keep cool. Avoid dehydration (b) Wear lightweight, light-coloured, loose, cotton clothes (c) Cover your head (d) Drink sufficient water- even if not thirsty (e) Use ORS, homemade drinks like lassi, torani (rice water), lemon water, buttermilk, etc. to keep yourself hydrated (f) Avoid alcohol, tea, coffee and carbonated soft drinks, which dehydrates the body (g) Take bath in cold water frequently. <u>In case of SUNSTROKE:</u> Lay the person in a cool place, under a shade. Wipe her/him with a wet cloth/wash the body frequently. Pour normal temperature water on the head. The main thing is to bring down the body temperature. Consult a Doctor immediately.
Red Alert (Take Action)	Extreme Heat Alert for the day	(i) Severe heat wave persists for more than 2 days. (ii) Total number of heat/severe heat wave days exceeding 6 days.	Very high likelihood of developing heat illness and heat stroke in all ages.	Along with suggested action for Orange Alert, Extreme care needed for vulnerable people.

(Source: National Guidelines for Preparation of Action Plan- Prevention and Management of Heat wave)

Heatwave Alert Levels & Required Actions

GREEN - Normal Conditions

Routine preparedness activities as outlined in pre-season measures: checking fans, coolers, water supply, PHC readiness, training healthcare and field workers, stocking ORS/IV fluids, and reviewing last five years temperature and humidity records.

YELLOW - Heatwave Alert (Be Updated)

Actions triggered when isolated pockets report heatwave for 2 days: public advisories, early warning dissemination by IMD through DEOC, adjusting school hours, ensuring uninterrupted power supply, rescheduling labour shifts, and strengthening PHC readiness.

ORANGE - Severe Heatwave Alert (Be Prepared)

Triggered when severe heatwave persists for 2 days or heatwave persists 4+ days: activation of DEOC, cooling centres, water tankers, medical team deployment, restricted permissions for public gatherings, and mandatory hospital reporting of heat illnesses.

RED - Extreme Heatwave Alert (Take Action)

Highest priority response when severe heatwave persists for more than two days or total heatwave days exceed 6: district-wide emergency activation, daily reporting, mandatory hydration and rest for outdoor workers, halting large events, ensuring no power cuts in critical facilities, and rapid heatstroke case referral.

4. Roles & Responsibilities (Operational Summary)

- District Collector (Incident Commander): Issues alerts, chairs coordination meetings, authorises activation of cooling centres and emergency measures.
- DDMA / DEOC: Monitors IMD forecasts, disseminates alerts, compiles daily reports, coordinates with all departments, and maintains communication channels.
- Health Department: Prepares Cold Wards, ensures ORS/IV fluids availability, trains staff in heatstroke management, activates Rapid Response Teams, and records/report heat illness cases.
- Police & Traffic: Manages crowd movement, provides shade at traffic booths, restricts public events in peak hours, and assists in emergency response.

- Municipal Corporations / ZP: Ensures drinking water, fans, cooling spaces, park access, shade in public places, and sanitation support during alerts.
- Labour Department: Implements shift changes, hydration facilities, rest breaks, and monitors working conditions for outdoor workers.
- School Education: Modifies school timings, restricts outdoor activities, ensures hydration and functioning fans in schools.

5. Heatstroke Case Management SOP

- Identify: high body temperature, hot dry skin, confusion, fainting, no sweating- symptoms matching Heatstroke definition in the treatment protocol.
- Immediate Action: move patient to shade/cool area, use wet cloth, fan, spray water, apply ice packs at neck/armpits/groin; avoid oral fluids if unconscious.
- Referral: call 108/102, activate ambulance, pre-alert nearest PHC/CHC/DH, maintain heatstroke register, and record cause appropriately.

6. Communication SOP

All heatwave alerts must follow the IMD → Collector → DEOC → Line Departments → Public dissemination chain, as required under the Early Warning System guidelines.

- Channels: SMS, WhatsApp, press notes, radio, local TV, Panchayat loudspeakers, social media, municipal alert systems.
- Reporting: Yellow- weekly; Orange- daily; Red- twice daily; format aligned with Annexure 5 reporting mechanisms.

7. Post-Heatwave Actions

Following the heatwave period, the district shall evaluate the response, review morbidity/mortality data, conduct interdepartmental meetings, and revise the action plan for the next season, consistent with the monitoring and evaluation framework described in Chapter 9.

Chapter 5: Measures for Prevention and Mitigation of the Effects of Heatwave

The number of Heatwaves events are rising and will become a major hazard for the District of Nashik in the coming future. But there are several things that individuals, community and the state can do to prevent and mitigate the harmful effects of Heatwave. Here are some ways through which the impacts of Heatwave can be reduced:

For Individuals and Community:

1. **Be Prepared at Home:** The best time to prepare for Heatwave is during Springs, i.e., the months of February to April. Each spring, check the fans, air conditioner and other cooling equipment in the household. Also do create a list of vulnerable family members, friends or neighbours and make sure their phone numbers are present. You should be aware about the nearest hospital.
2. **Be prepared in Community:** Make sure any event that takes place during the Heatwave period happens during the coolest period of the day, preferably, early morning. People involved in a physically demanding job without air conditioning should establish a committee at your workplace to develop a heat response plan.
3. **Install a Cool Roof:** A cool roof consists of materials or coatings designed to deflect sunlight and heat away from residential structures, thereby lowering roof temperatures. This results in a cooler indoor environment, enhancing overall comfort and diminishing the necessity for excessive air conditioning usage on hot days.

Furthermore, cool roofs present a straightforward and economically viable response to challenges posed by urbanization. By conserving energy, enhancing thermal comfort, and decreasing the demand for cooling, they contribute to sustainable urban development. The reflective nature of cool roofs enables them to deflect sunlight and minimize heat absorption. Depending on the context, the implementation of cool roofs may lead to a reduction in indoor temperatures ranging from 2 to 5 degrees Celsius (comparison to conventional roofing systems).

4. **Planting of trees or the installation of shade structures strategically:** Incorporating trees and vegetation that provide direct shading for your residence can effectively lower ambient temperatures, subsequently reducing the reliance on air conditioning,

enhancing home comfort, and mitigating energy expenditures. Beyond these advantages, trees contribute to the well-being of your family by improving air quality, offering cooling shade for outdoor activities, and diminishing sun exposure.

5. **Utilize cool paving materials for your driveway:** The rapid heating of pavement under sunlight is a familiar experience, particularly when walking barefoot. Not only does hot pavement add to the urban heat island effect by transferring heat to the surrounding air, but it also escalates discomfort. Opting for cool pavement materials, such as those made from asphalt and concrete, or incorporating coatings and grass paving, ensures that the surface remains cooler in the sun. This is achieved through the reflection of solar energy or the promotion of water evaporation.
6. **Upgrade outdated cooling devices:** Replace inefficient and energy-consuming cooling units that are outdated or damaged.
7. **Familiarize yourself with local heat alert systems and subscribe to them:** Explore the existence of heat alert systems established by local governments, weather stations, medical providers, or other relevant entities. Install the ‘SACHET’ application to receive weather warnings.

For District:

1. **Early Warning Systems:** Implement a robust early warning system to forecast and communicate impending heatwaves, enabling residents and authorities to take proactive measures.
2. **Public Awareness Campaigns:** Conduct comprehensive public awareness campaigns to educate the public on the risks associated with heatwaves and promote preventive measures, such as staying hydrated, seeking shade, and using cooling centres
3. **Cooling Centres and Shelters:** Establish and maintain cooling centres across urban and rural areas, especially in densely populated regions. Ensure these centres are well-equipped with cooling facilities, water, and medical support.
4. **Community Engagement:** Foster community engagement programs that encourage neighbours to check on vulnerable individuals, particularly the elderly, those with chronic illnesses, and people living alone, during periods of extreme heat.

5. **Urban Planning and Green Spaces:** Integrate heat-resilient urban planning practices, including the creation of green spaces and the planting of trees to mitigate the urban heat island effect and provide cooler areas for residents.
6. **Workplace Guidelines:** Collaborate with industries and businesses to develop and implement heatwave guidelines for workplaces. These guidelines should prioritize worker safety, including the provision of shaded rest areas and adequate hydration facilities. Encourage employers to adopt heat safety guidelines for outdoor workers. Implement flexible working hours or remote work options during extreme heat events.
7. **Infrastructure Resilience:** Enhance the resilience of critical infrastructure, such as power grids and water supply systems, to withstand the increased demands and potential failures during heatwaves.
8. **Emergency Response Preparedness:** Develop and regularly test emergency response plans that outline specific actions to be taken by various agencies and stakeholders during a heatwave. This includes coordination with healthcare facilities, emergency services, and local authorities.
9. **Healthcare Capacity Building:** Strengthen healthcare systems by providing additional resources, training, and capacity-building programs for medical professionals to handle an increased number of heat-related illnesses during extreme weather events.
10. **Energy Conservation Programs:** Collaborate with the energy sector to implement and promote energy conservation programs, reducing overall energy demand during peak periods and minimizing the risk of power outages.
11. **Water Management Strategies:** Implement water management strategies to ensure a sustainable and reliable water supply during heatwaves. This includes measures to reduce water loss, enhance water conservation, and manage water distribution efficiently.
12. **Research and Data Collection:** Support research initiatives to gather localized data on heatwave patterns, vulnerabilities, and community-specific impacts. This information can inform targeted interventions and adaptation strategies.

13. **Legislation and Regulations:** Enforce and strengthen regulations related to urban planning, construction practices, and energy efficiency standards to enhance resilience against heatwaves.
14. **Collaboration with NGOs and Civil Society:** Collaborate with non-governmental organizations (NGOs) and civil society groups to enhance community resilience, provide support to vulnerable populations, and facilitate grassroots initiatives for heatwave preparedness.
15. **Cross-Sectoral Coordination:** Facilitate coordination and communication among different sectors, including health, agriculture, education, and transportation, to ensure a comprehensive and integrated approach to heatwave prevention and mitigation.

Prevention of Heat Related Illness

In the context of Nashik, India, prolonged and intense heat waves emerge as a primary contributor to heightened rates of illness and mortality. The impact of extreme summer temperatures on human well-being is further intensified by elevated humidity levels. Compelling evidence suggests that the mortality impact of heat waves is particularly pronounced on days marked by elevated levels of ozone and fine particulate matter. The impending effects of global climate change pose a concerning outlook for Nashik, projecting an escalation in the frequency, intensity, and duration of heat waves, along with a corresponding rise in attributable deaths, as outlined by the World Health Organization (WHO).

Mitigating heat-related illnesses is within reach. Empowering vulnerable populations and communities in Nashik with awareness regarding prevention strategies, essential dos and don'ts, disseminated through diverse media channels, stands as a crucial preventive measure. Equipping physicians and pharmacists with knowledge of effective prevention techniques, along with first-aid treatment, is imperative. Additionally, fostering awareness about the potential side effects of prescription drugs in hot weather is vital for healthcare professionals to effectively address and alleviate the impact of heat-related illnesses in the region. The table below shows the symptoms and first aid for various Heat Disorders.

Table 5: Symptoms and First Aid for various Heat Disorders.

Heat Disorder	Symptoms	First Aid
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Heat rash	Skin redness and pain, possible swelling, blisters, fever, headaches.	Take a shower using soap to remove oils that may block pores preventing the body from cooling naturally. If blisters occur, apply dry, sterile dressing and seek medical attention.
Heat Cramps	Painful spasms usually in leg and abdominal muscles or extremities. Heavy sweating.	Move to cool or shaded place. Apply firm pressure on cramping muscles or gently massage to relieve spasm. Give sips of water. If nausea occurs, discontinue.
Heat Exhaustion	Heavy sweating, weakness, Skin cold, pale, headache and clammy extremities. Weak pulse. Normal temperature possible. Fainting, vomiting.	Get victim to lie down in a cool place. Loosen clothing. Apply cool, wet cloth. Fan or move victim to air-conditioned place. Give sips of water slowly and if nausea occurs, discontinue. If vomiting occurs, seek immediate medical attention, call 108 and 102 for ambulance.
Heat Stroke (Sun Stroke)	High body temperature. Hot, dry skin. Rapid, strong pulse. Possible unconsciousness or altered mental status. Victim will likely not sweat.	Heat stroke is a severe medical emergency. Call 108 and 102 for ambulance for emergency medical services or take the victim to a hospital immediately. Delay can be fatal. Move victim to a cooler environment. Try spraying water, cold water on body & fan the wet body. If possible sponging or cool bath sponging to reduce body temperature. Use extreme caution. Use fan and/or conditioners. DO NOT GIVE FLUIDS ORALLY if the person is not conscious.

(Source: National Guidelines for Preparation of Action Plan- Prevention and Management of Heat wave)

Annexure 1: District and Other Control Room Numbers

S.No.	Office/Official	Name	Mobile	Office No./ Email ID
District Control Room				
1	District Collector & Chairman	Shri Ayush Prasad	9845568917	0253 2578500 collector.nashik@maharashtra.gov.in
2	Resident Deputy Collector & CEO	Shri. Rohitkumar Rajput	8454072637	0253 2578501 rdcnashik@gmail.com
3	District Emergency Operation Centre- DDMO	Srikrishna Deshpande	7387442580	0253-2317151 ddmanashik@gmail.com
Toll Free No: 1077				
Police				
1	City Police Control Room	Nashik - City		112 0253-2305233/34
2	District Police Control Room	Nashik Rural		0253-2200401
3	Superintendent of Police (SP)	Shri. Balasaheb Patil (IPS)	8652781777	sp.nashik.r@mahapolice.gov.in
4	Commissioner of Police (CP)	Sandeep Karnik (IPS)	9823126333	0253-2305200 cp.nashik@mahapolice.gov.in
NMC				
1	Additional Commissioner DMR&R - NMC	Smita Zagade	9850275557	

2	Municipal Disaster Control Room Number -	Nashik		0253-2571872 0253-2317505 0253-2222413
Fire Brigade NMC / MMC				
1	Chief Fire Officer (NMC)	Shri. Bairagi	9420690443	0253-2590871
2	Chief Fire Officer (MMC)	Shri. Pawar	9823339275	02554-234567
Civil Defence / Homeguards				
1	District Civil Defence Control Room	Athul Shivalal Jagtap	8550966801	0253-2573149 dccd.nashik@gmail.com
2	District Home guard Control Room	Suresh Jadhav	9421507632	0253-2578346 dchgnashik@yahoo.com
Health				
1	District Health Officer	Dr. Sudhakar More	7588930338	dhonashik@rediffmail.com
2	District Civil Surgeon	Dr. Charudatt Shinde	8108932111	csnashiksinhastha2015@gmail.com
3	Medical Officer of Health (NMC)	Dr Vijay Devkar	9922020306	nmcmsmd@gmail.com
4	Medical Officer of Health (MMC)	Dr Jayashri Aher	9422906989	mlghealth@gmail.com

Annexure 2: Contact Number of Health Department

Contact of Nodal Officers

S. No.	Department	Name of Nodal Officer	Contact Number	Email
1	Health	Dr.Rajendra Bagul	9960912303	idspnasik1@gmail.com
S. No.	District	Name of Nodal Officer (THO)	Contact Number	Email
1	Baglan	Dr.Harshalkumar Mahajan	9022973752	tmobaglan2020@gmail.com
2	Chandwad	Dr. Hemant Ahhirao	8275021446	thochandwad3@gmail.com
3	Deola	Dr.Sudhir Patil	7774048483	thodeolanrhm@gmail.com
4	Dindori	Dr.Kalpesh Chopade	9970849796	tmodindori1@gmail.com
5	Igatpuri	Dr.Deshmukh	9890620786	tmo.igatpuri@gmail.com
6	Kalwan	Dr.Bangal	9890389894	kalwantmo@gmail.com
7	Malegaon	Dr.Akash Bhadre	7507281724	tmomalegaon@gmail.com
8	Nandgaon	Dr.Santosh Jagtap	9423324262	tmonandgaon@gmail.com
9	Nashik	Dr.Shailesh Nikam	9423972131	thonashik1@gmail.com
10	Niphad	Dr.Sujit Koshire	9422757565	tmoniphad@gmail.com
11	Peth	Dr.Yogesh More	9665121311	tho.pspeth2017@gmail.com
12	Sinnar	Dr.Mohan Bachav	7875411711	tho.sinnar@gmail.com
13	Surgana	Dr.Ranveer	9370734521	tho.surgana123@gmail.com
14	Trimbak	Dr.Motilal Patil	9850311539	tmo.trimbak123@gmail.com
15	Yeola	Dr.Sharad Katkade	9921594048	thoyeola@gmail.com

Contact of Medical Institutes

District Hospitals				
S. No	Name of Hospital	Block	Name of Nodal Officer	Telephone
1	District Hospital Nashik	Nashik	Dr.Nilesh Patil	9699155145
2			Dr.Pradnya Ahire	8975509998
3			Dr.Suhas Patil	9767598421
4			Dr.Pratik Bhangare	8007100717
5			Smt.Anita Bhalerao	8975702336
Rural Institutes				
S. No	Name of Institute (RH/SDH)	Block	Name of Nodal Officer (SDH/RH)	Telephone
1	RH Dangsaundane	Satana	Dr.Manojkumar B.Gadekar	7588515800
2	RH Nampur	Satana	Dr.Mahendra Patil	7721878877
3	RH Satana	Satana	Dr.Manojkumar Gadekar	7588515800
4	SDH Chandvad	Chandwad	Dr.Navnath E.Avhad	8999238120
5	RH Deola	Deola	Dr.Yogesh Ahire	9422770229
6	RH Umrane	Deola	Dr.Navnath E.Avhad	8999238120
7	RH Dindori	Dindori	Dr.Hemantkumar Ghangal	9421510988
8	RH Vani	Dindori	Dr.Sonali Gaydhani	9323792673
9	RH Ghoti	Igatpuri	Dr.Rahul Wagh	9881915657
10	RH Igatpuri	Igatpuri	Dr.Pradip Ingole	9130733202
11	RH Abhona	Kalwan	Dr.Vandana P.Chaure	7350294037
12	SDH Kalwan	Kalwan	Dr.Hemant Pawar	9011776902 7977590588
13	RH Dabhadi	Malegaon	Dr.Dnyaneshwari S.Dange-Pawar	8208835562
14	RH Zodga	Malegaon	Dr.Pramod K. Khairnar	9673164555
15	GH Malegaon	Malegaon	Dr.Laxman N.Chavan	9860295207
16	WCH Malegaon	Malegaon	Dr.Dipak K.Pawar	7030164570
17	RH Nandgaon	Nandgaon	Dr.Kajal Salunkhe	7020074849
18	SDH Manmad	Nandgaon	Dr.Jayesh More	9960786729
19	RH Girnare	Nashik	Dr.Prashant Deore	9422759823
20	RH Lasalgaon	Niphad	Dr Balkrishna Ahire	9819359142
21	RH Pimpalgaon Basvant	Niphad	Dr.Nagnanda Magare	9623450833
22	SDH Niphad	Niphad	Dr.Rajendra Tambhale	9822802397
23	RH Peint	Peth	Dr.Samdhan Patil	9421520666
24	RH Dodi	Sinnar	Dr.Kalyani S.Vasnik	8975423742
25	RH Sinnar	Sinnar	Dr.Nirmala N.Gaikwad	89751 51355
26	RH Barhe	Surgana	Dr.Rajendra C.Borase	8446661719
27	RH Surgana	Surgana	Dr.Sanjay C.Chaudhari	9011095728
28	RH Harsul	Trimbak	Dr.Pramod Gunjal	9011928067
29	SDH Trimbak	Trimbak	Dr.Ramesh Pawar	9420335958

30	RH Nagarsul	Yeola	Dr.Sandip Suryavanshi	9403566647
31	SDH Yeola	Yeola	Dr.Ramkant G.Sonawane	8857876400

Contact Number of Primary Health Centers (PHC)

S. No	Name of PHC	Block	Name of Nodal Officer (MO)	Telephone
1	Ambasan	Baglan	Dr. Rahul Sharad More	9763438970
2	Bramhangaon	Baglan	Dr. Prajkta Jalindhar Banga	8329945928
3	Jaikheda	Baglan	Dr.Julesh Sonawane	8390117073
4	Kapaleshwar TR	Baglan	Dr. Yogeshwari jibhau gaikwad	9423563251
5	Kelzar TR	Baglan	Dr.Rahul Gayakawad	9764648285
6	Mulher TR	Baglan	Dr.Mahendra More	9623386402
7	Nirpur	Baglan	DR. Rahul vasantrao sonawane	7040001111
8	Salher TR	Baglan	Dr.Abhijit Nakul Sonje	9960177565
9	Taharabad	Baglan	Dr. Anita Sonawane	9657715832
10	Virgaon	Baglan	Dr.Nilesh Deore	9545515074
11	Kazisangvi	Chandvad	Dr.Suyash Sonwane	9421407078
12	Talegaon Rohi	Chandvad	Dr.Renuka Ghonge	9766390817
13	Uswad	Chandvad	Dr.Rutuja Tompe	7887447709
14	Vadalibhoi	Chandvad	Dr.Yogini Thakare	9423024631
15	Vadner Bhairao	Chandvad	Dr.Gaurav Kulkarni	8856800626
16	Dahiwad TR	Deola	Dr. Laxmikant Baradkar	8010429450
17	Khamkheda TR	Deola	Dr. Nilesh Thakare	9860163222
18	Kharda TR	Deola	Dr. Santoshkumar Ade	9422650726
19	Lohoner	Deola	Dr. Sonam Vishwkarma	7420843160
20	Meshi	Deola	Dr. Sandip Pawar	9511622859
21	Khedgaon	Dindori	Dr Bandu Ghodsare	9503480044
22	Kochargaon TR	Dindori	Dr Trerashami Bukane	7972297626
23	Mohadi TR	Dindori	Dr S B Bairagi	9421602519
24	Nanashi TR	Dindori	Dr P U Joshi	8208683448
25	Nigdol TR	Dindori	Dr Raosahab Khot	9822654879
26	Pandane TR	Dindori	Dr L S Sable	9623026126
27	Talegaon Dindori TR	Dindori	Dr Leena Bhalshankar	9850803873
28	Umrale Bk TR	Dindori	Dr Rekha Sonawane	8691872040

S. No	Name of PHC	Block	Name of Nodal Officer (MO)	Telephone
29	Vare TR	Dindori	Dr R G Jople	8108417371
30	Varkheda TR	Dindori	Dr D D Daule	8605214864
31	Belgaon Kurhe	Igatpuri	Dr Vaishali Thakare	7666803207
32	Dhamangaon	Igatpuri	Dr Vasaikar	9405792146
33	Kaluste TR	Igatpuri	Dr. Navale	8805235207
34	Kananwadi TR	Igatpuri	Dr Ragini Gangurde	9420286320
35	Khed TR	Igatpuri	Dr Chitra Vedhe	9765719159
36	Nandgaonsado TR	Igatpuri	Dr Prataprao B Deshmukh	8805993619
37	Vaitarana TR	Igatpuri	Dr Lachake B. S	9552243013
38	Wadivarhe	Igatpuri	Dr Dipika More	7030469365
39	Dalwat TR	Kalwan	Dr.Sheshikant Ubale	9527824323
40	Jaidar TR	Kalwan	Dr.Ashey Giri	7972816148
41	Kanashi TR	Kalwan	Dr.Eshan Dongarde	8308153747
42	Mokbhangi TR	Kalwan	Dr.Sudeshna Bagul	8888962832
43	Nanduri	Kalwan	Dr.Jayshri Patil	9689293781
44	Navi Bej TR	Kalwan	Dr.Nitin Mahajan	9527264530
45	Otur TR	Kalwan	Dr.Prashant Bahiram	9834168001
46	Tirhal Kh TR	Kalwan	Dr.Virendra Avari	9822928852
47	Umbarghavan	Kalwan	Dr.Dhananjay Devre	9604828696
48	Chikhalohol	Malegaon	Dr Vikram Shantaram Jadhav	8275129881
49	Kalwadi	Malegaon	Dr Poonam Manikrao Jadhvar	7083622553
50	Karanjgavhan	Malegaon	Dr Vasumati Laxman Shewale	8975062133
51	Malgaon	Malegaon	Dr Mohnish Sahebrao Raundal	7588515494
52	Nimgaon	Malegaon	Dr Madhuri Nandu Shinde	8263038506
53	Ravalgaon	Malegaon	Dr Amol Ramanlal Jadhav	9421907560
54	Saundane	Malegaon	Dr Aishwarya Sanjay Panpalia	9922613819
55	Sonaj	Malegaon	Dr Kureshi Jiyaurrehman	8928665556
56	Vadner Khakurdi	Malegaon	Dr Dipak Laxman Nikam	9890582557
57	Bolthan	Nandgaon	Dr.Swati Madhraov Gond	9673472822
58	Hiswal	Nandgaon	Dr.Vinaya Suresh Borase	9423324263
59	Naydongri	Nandgaon	Dr.Ritikesh Vasudev Mali	9765078446

S. No	Name of PHC	Block	Name of Nodal Officer (MO)	Telephone
60	Pimparkhed	Nandgaon	Dr.Amit Yashavant Gaikwad	9096904296
61	Vehelgaon	Nandgaon	Dr.Jitesh Arvind Patil	9503389893
62	Dhondegaon TR	Nashik	Dr. Amit Patil	9422802508
63	Jategaon TR	Nashik	Dr. pradeep jaybhawe	9860566000
64	Lahavit	Nashik	Dr. jaymala ramsingh vasave	9270000082
65	Sayyad Pimpari	Nashik	Dr. Madhav Ahire	9922119799
66	Shinde	Nashik	Dr. Nitin Badade	9975650751
67	Chandori	Niphad	Dr Enjal chavan	7758805330
68	Deogaon	Niphad	Dr Mayur Barbande	8800771024
69	Kasabe Sukene	Niphad	Dr Nisha Kothe	8698399920
70	Khadak malegaon	Niphad	Dr Kalyan shinde	9373315693
71	Mhalsakore	Niphad	Dr Sanjay Dani	9881258800
72	Naitale	Niphad	Dr P B Charode	9422728153
73	Nimgaon Wakada	Niphad	Dr Varunraj Thakare	8956423510
74	Ozar	Niphad	Dr Prachi pawar	8169465070
75	Palkhed	Niphad	Dr Rahul Khatal	7666468331
76	Pimpalgaon Baswant	Niphad	Dr Yogesh Dhanwate	9923741149
77	Ambe TR	Peth	Dr.Kothavade Hemant Bhalchandra	9403418077
78	Bhuwan TR	Peth	Dr.Bagul Prashant Devram	9422903509
79	Jogmodi TR	Peth	Dr.Patel Rohini Madhukar	9307795941
80	Karanjali TR	Peth	Dr.Sarvade Durgadas Dattatray	9024937238
81	Kohor TR	Peth	Dr.Bunge Kalyani Popatrao	9673477677
82	Kulwandi TR	Peth	Dr.Devre Atul Rajendra	9423184429
83	Kumbhale TR	Peth	Dr.Shinde Sushma Keshav	8767985014
84	Dapur	Sinnar	Dr.Dhiraj Kedar	7720968592
85	Devpur	Sinnar	Dr.Kamini Shinde	9049205862
86	Naigaon	Sinnar	Dr.Manisha Wagh	8329517497
87	Pandhurli	Sinnar	Dr.Lahu Patil	9881779151
88	Somthane	Sinnar	Dr.Akshay Kanawade	9561615999
89	Thangaon	Sinnar	Dr.Gayatri Shirsul	8830531270

S. No	Name of PHC	Block	Name of Nodal Officer (MO)	Telephone
90	Wavi	Sinnar	Dr.Shital Roy	9404703826
91	Barhe TR	Surgana	Dr.Apeksha Tukaram Jadhav	9960639474
92	Borgaon TR	Surgana	Dr Harshada Sanap	9029139034
93	Bubli TR	Surgana	Dr. Pradip Raut	9604892182
94	Mani TR	Surgana	Dr.Sagar Sonaji Gavali	9860627480
95	Mankhed TR	Surgana	Dr.Vijaykumar Ramu Sathe	7057034808
96	Palsan TR	Surgana	Dr.Surekha Manicharan Lohat	9405894707
97	Pangarne TR	Surgana	Dr.Jayendra Chandrakant Thavil	8624011363
98	Umbarthan TR	Surgana	Dr.Rakesh Subhas Sonwane	9403512088
99	Amboli TR	Trimbakeshwar	Dr Pankaj Wadile	9420439629
100	Anjaneri	Trimbakeshwar	Dr Ashish Sonwane	9822826827
101	Chinch Ohol TR	Trimbakeshwar	Dr Ajay Kapse	9922424080
102	Mulvad TR	Trimbakeshwar	Dr Hirshi Paik	7743871751
103	Rohile TR	Trimbakeshwar	Dr Gautam Rajguru	8080105275
104	Shirasgaon TR	Trimbakeshwar	Dr Dyneshwar Navale	8605462411
105	Thanapada TR	Trimbakeshwar	Dr Sangita Pawar	7020302674
106	Andrsul	Yevla	Dr. Jayshree Pawar	8390485541
107	Bharam	Yevla	Dr. Akshay Navgire	8007555485
108	Mukhed	Yevla	Dr. Tushar Baviskar	9421930168
109	Patoda	Yevla	Dr. Tejas Jangam	7887303813
110	Rajapur	Yevla	Dr. Sachin Kutte	7020163883
111	Savargaon	Yevla	Dr. Sonali Gattuwar	7058052695

Annexure 3: Contact Number of Tehsildars

Sr.No.	Name of Officers	Tehsil	Mobile No.
1	Mr. Abasaheb Tambe	Tehsildar (Revenue)	9503921997
2	Mr. Pradip Varpe	Tehsildar, (General)	90965 41097
Total Tehsil Offices -15			
3	Mr. Milind Kulkarni	Tehsildar Deola	9552564041
4	Mr. Vishal Sonwane	Tehsildar Malegaon	9423666765
5	Mr. Pankaj Pawar	Tehsildar, Nashik	9922590653
6	Mr. Kailas Chavde	Tehsildar Bagalan	9960242166
7	Mr. Mukesh Kamble	Tehsildar Dindori	9226055647
8	Mr. Vishal Naikwade	Tehsildar Niphad	7276781386
9	Mr. Sunil Saindane	Tehsildar Nandgaon	9422769440
10	Mr. Ganesh Digambar Jadhav	Tehsildar Trimbakeshwar	7499977101
11	Mr. Anil Ratilal Chavan	Tehsildar Chandwad	8380051231
12	Mr. Rohidas Warule	Tehsildar Kalwan	8308334457
13	Mr. Abhijit Baravkar	Tehsildar Igatpuri	9890838498
14	Mr. Aba Mahajan	Tehsildar Yeola	9423492088
15	Mr. Ramji Rathod	Tehsildar Surgana	7741014112
16	Mr. Surendra Deshmukh	Tehsildar Sinnar	7350560501
17	Mr. Shivaji Madhukar Gawli	Tehsildar Peth	8390387206

Annexure 4: Contact Number of CP and SP

List of names and phone no of officers from the Police Commissionerate			
Name of Police Officer	Designation	Office No.	Mobile No.
Mr. Sandeep Karnik	Police Commissioner Nashik City	2305200/	9823126333
Mr. Sanjay Kankhar	Personal Assistant	2305201	7028001650
Mr. Dharmaraj Bangar	Asst. Police Inspector (Petitioner)	2305227	9975393441
Mr. Chandrakant Khandvi	Deputy Commissioner of Police, Headquarters	2305205	9822634500
Shri. Kiran Kumar Chavan	Deputy Commissioner of Police, Headquarters, Zone-1	2305207	9764347062
Smt. Monica Raut	Deputy Commissioner of Police, Headquarters, Zone-2	2503431	8424000689
Mr. Prashant Bachhao	Deputy Commissioner of Police, (Crime /SB)	2305209	9823417764
Smt. Sangita Nikam	Asst. Commissioner of Police, Administration	2305211	7798530397
Mr. Sandeep Mitke	Asst. Commissioner of Police, Crime Branch	2305213	8149532577

Mr. Sandeep Mitke	Asst. Commissioner of Police, Special Branch	2305246	8149532577
Smt. Padmaja Bade	Asst. Commissioner of Police, Panchvati	2305215	9657689595
Mr. Nitin Jadhav	Asst. Commissioner of Police, Sarkar wada	2315985	9890944934
Mr. Shekhar Deshmukh	Asst. Commissioner of Police, Ambad	2310119	8766838401
Mr. Sachin Bari	Asst. Commissioner of Police, Nashik Road	2461233	9552273100/ 9049872822
Mr. Sudhakar Suradkar	Asst. Commissioner of Police, Traffic	2305228	9370059020
Mr. Nagesh Hadpe	Administrative Officer	2952218	9823732999
Mr. Vijay Londhe	Public Relations Officer	2305249	9823120402
Mr. Sachin Wakade	Police Sub-Inspector, Reader-2		9623003273

List of names and phone no of Superintendent of Police, Nashik Rural Officers

Police Station	Name of Officers	Office No	Mobile No
Superintendent of Police	Shri. Balasaheb Patil (IPS)	0253-2309700	8652781777
Additional Superintendent of Police, Malegaon	Tegbir Singh Sandhu (IPS)	02554-222022	9953198978
Additional Superintendent of Police, Nashik Rural	Aditya Mirkhelkar (IPS)	0253-2309722	8087673440

Deputy Superintendent of Police (HQ)	Nitin Kumar Gokave	0253-2309702	9823020644
Deputy Superintendent of Police, EOW	Mrs. Karandikar	0253-2200422	8459392560
Sub Divisional Officer, Nashik, Rural	Harish Khedkar	0253-2309713	8693031333
Igatpuri	Sarika Ahirrao	02553-244004	8975587100
Ghoti	Vijay Shinde	02553-220544	9923879951
Wadivare	Bagavan Mathura	02553-236533	8888745745
Ozar	Nitin Kandare	02550-278633	9822908062
Pimpalgaon	Pradeep Deshmukh	02550-250033	8010114411
Sub Divisional Officer, Niphad	K. K Patil	02550-241053	9823515373
Niphad	Ganesh Gurav	02550-241033	9527233777
Sinnar	Hemant Kumar Bhamre	02551-220033	8983001009
Lasalgaon	Bhaskar Shinde	02550-266055	9689888618
Wavi	Ganesh Shinde	02551-283433	9702231818
Saykheda	Sandeep Ghuge	02550-232233	8605111234
Sinner MIDC	Bharat Jadhav	02551-240433	9423230133
Sub Divisional Officer, Peth	Vasudev Desale	02594-233144	9823101778
Nashik Thaluk	Mridula Naik	0253-2309711	9881355983
Peth	Dwarkanath Gondke	02558-225533	9823777517
Trimbakeshwar	Mahesh Kulkarni	02594-233133	8975020400

Harsul	Jeevan Borse	02558-227233	8380089711
Sub Divisional Officer , Kalwan	Kiran Kumar Suryavanshi	02592-250033	9975631374
Kalwan	Khagendra Tembhekar	02592-221033	9823138949
Dindori	Raghunath Shegar	02557-221033	9921216277
Wani	Gayathri Jafhav	02557-220133	7588516042
Abona	Srimathi Kokate	02592-240033	9823322505
Surgana	Vinod Patil	02593-223333	9823724385
Deola	Sarthak Nehte	02592-228233	9422666775
Ba- He	Pradeep Gite	7588977564	9321658725
Sub Divisional Officer, Manmad	Bajirao Mahajan	02591-222254	7272047979
Manmad	Vijay Kare	02591-222920	8459943611
Yeola Taluk	Sandeep Mandlik	02559-265073	9403689456
Yeola City	Anil Bhawari	02559-265016	9004662127
Nandgaon	Digambar Bhadane	02552-242333	8108511811
Chandwad	Kailas Wag	02556-252233	9821839899
Sub Divisional Officer, Malegaon	Siddeshwar Akhegaonkar	02556-275633	9960937805
Malegaon Taluk	Baviskar	02554-256500	9923694251
Satana	Pritham Chaudhary	02554-276162	9867716103
Vadner Khakudi	Yogesh Patil	02555-223033	9823360886
Jaikheda	Purushottam Shirsath	02554-277833	9892852336
	Sagar Kale	02555-233433	8888181777

Sub Divisional Officer, Malegaon City	Siddharth Barwal Bhapose	02554-222004	8400197213
Malegaon City	Sandeep Patil	02554-222001	9850069584
Azadnagar	Yogesh Ghorpade	02554-222006	9870273727
Ayeshanagar	Gautam Tayde	02554-222009	9763322358
Pawarwadi	Rahul Khatal	02554-280100	7977406198
Ramjanpura	Ananda Mali	-----	8108666923
Sub Divisional Officer, Malegaon Camp	Darshan Dugad Bhapose	02554-258901	7249636536
Malegaon Camp	Deepak Patil	02554-258904	9272735529
Killa	Sudir Patil	02554-222005	8888305501
Malegaon Chavani	Deepak Jadhav	02554-258906	9420371253

Annexure 5: Detailed Phase-wise Roles and Responsibilities of various departments

S. No.	Name of the Department	Responsibilities		
		Phase 1: Pre-Heat Wave Season (January to April)	Phase 2: During the Heat Wave Season (May to August)	Phase 3: Post -Heat Wave Season (September to December)
1.	District Disaster Management Authority	• Preparation of a list of High-risk areas in the State vulnerable to Heat waves. • Identification of vulnerable groups of the population. • Convene meetings with the concerned Departments/ Agencies/ NGOs involved in the response mechanism to review the action plan periodically. • Designation of a single officer as a point of contact for each department. • Organize training for health workers, line workers, school children, and the local community in preventive measures and treatment protocol	• Issue of heat alert when extreme heat events are forecasted by IMD. All key Departments / Agencies, SEOC, DEOC, etc. in accordance with the Communication Plan may be notified. Periodicity depends upon the severity of the forecast. • Monitor and increase the heat alert level to match the severity of the forecast and established threshold. • Hold regular (daily, if necessary) conferences to discuss reports and fresh developments during a heat alert. Special meetings with key agencies may be convened.	• Review of quantitative and qualitative data for process evaluation and improvements. • Annual evaluation of the Heat Wave Action Plan by organizing a meeting with key departments /agencies and relevant stakeholders. • Evaluate the Plan process based on the reach and impact. Revision of the Plan based on the performance feedback.

		involving the Medical & Health Department. • Distribute pamphlets and posters with tips to prevent heat stress in the local language to hospitals, schools, and professional associations. • Establish a Heat Wave Action Web Page on the Disaster Management / District Website	• To ensure that communication channels with all Stakeholders are functional and operating. • Ensure the presence of staff and availability of required supplies with each Department, including fresh drinking water. • Communicate locations of emergency facilities and cooling centers / shaded areas with each Department / Organization. • Inform power supply, Companies to prioritize maintaining power to critical facilities (such as hospitals and UHCs). • Notify all the stakeholders when the heat alert is over.	• Revision and posting of the Revised Action Plan online well ahead of summer season next year for information of all stakeholders.
2.	Information and Public Relation	• Identification of areas to post warnings and information during heat wave season.	• Release of messages to the general public and vulnerable groups about the risks and dangers of heat-related illness by the nodal officer at the	• Evaluate the reach of advertising / public messages and other means of communication like

		• Securing advertisement / scrolling slots for announcements regarding Heat waves. • Designing information and awareness material in the form of pamphlets, posters, etc. on Heat waves in the local language for distribution to the public, especially focusing on identified high-risk areas in the State and vulnerable groups of the population.	State and District levels through press conferences. • Wide circulation of Heatwave alerts through SMS or WhatsApp in collaboration with private sector Telecom companies in addition to traditional media during a heat alert. • Circulate heat alerts in bulk to the public via centralized email databases. • To send SMS alert messages directly to private practitioners in addition to the medical professionals at PHCs and UHCs. • Utilize local radio and FM broadcasts to disseminate heat protection tips and high-temperature warnings to vulnerable sections of populations.	• social media (Facebook, Twitter, etc.) to target groups. • Participate in the annual evaluation of the heat wave action plan. • Review the revised Heat Wave Action Plan.
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			Using social media like Twitter, Face book, etc. to increase the outreach of the messages.	
3.	Health and Family Welfare	- Designing and initiating targeted training programs, capacity-building efforts, and communication on heat illness for medical staff at Public Health Centres (PHCs) / local hospitals and Urban Health Centres (UHCs), including nursing staff, paramedics, field staff, and link workers (ANMs, ASHA Workers, etc.), while paying special attention to the susceptibility of wards. - Updating of admissions and emergency case records in Hospitals to track heat-related morbidity and mortality and to create simple, user-friendly means to track daily heat-related data and behavioural change impacts.	- Display heat-related illness prevention tips and how to stay cool around hospitals, PHCs, and UHCs. - Equip all hospitals/ PHCS/ UHCs with additional supplies of medicines and materials. Ensure the adoption of Heat illness treatment and prevention protocols. - Deploy additional staff at hospitals and PHCs/UHCs to attend to the influx of patients during a heat alert, if feasible. - Keep emergency wards ready in all PHCs / UHCs and Hospitals - Increase outreach of community health workers in at-risk neighbourhoods during a heat alert, if feasible.	- Evaluate the reach of advertising / public messages and other means of communication like social media (Facebook, Twitter, etc.) to target groups. - Participate in the annual evaluation of the heat wave action plan. - Review the revised Heat Wave Action Plan.

		• Train hospitals to record information on education & communication (IEC) efforts and to ensure the recording of the cause of death in death certificates. • Adopt heat-focused examination procedures at local hospitals and urban health centres. • Developing an SMS facility to reach the field level staff during emergencies. • Checking inventories of medical supplies including ORS powder in PHCs and other Local Hospitals. • Purchase and distribute reusable soft plastic ice packs for the citywide UHCs, 108 emergency centres, ambulances, and hospitals. • Explore creation of ice pack dispensaries to increase access to vulnerable communities in high-risk areas.	• Report Heatstroke patients to the nodal officer on a daily basis and generate weekly reports on public health impacts of Heatwave for the nodal officer, during a heat alert. • Expedite recording of cause of death in death certificates. • Ensure that regional health officers visit PHCs / UHCs to confirm that proper preparation has been made for heat-related illness and conduct case audits during heat season. • Ensure that 108 EMERGENCY SERVICE: • Activate a dynamic strategic deployment plan for ambulances. • Adequate supply of ice packs, IV fluids, and medicines. • Keep accurate records of pre-hospital care.	
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		• To provide the following services through 108 Emergency Service • Ensure adequate supply of IV fluids. • Prepare handouts for paramedics about heat-related illnesses. • Create displays on ambulances to build public awareness during major local events. • Identifying routes to high-risk areas and reaching vulnerable sections of the population in the shortest time possible by utilizing the list of high-risk areas.	• Adequate staff on duty and restrict leave if necessary.	
4.	Housing and Urban Development / Corporations / Municipalities / Urban Local Bodies	• High-Risk Area mapping and identification of vulnerable groups particularly destitute, homeless, beggars, and old age homes to concentrate on mitigation efforts during the heat alert period. • Identification of areas to provide shelters and drinking water during the heat alert period.	• Disseminate SMS text messages to warn residents of the high-risk areas and vulnerable sections of the population during a heat alert. • Set up electronic scrolling boards to display temperature and forecasts at junctions and other public places. • Activate “cooling centers,” such as public buildings, malls, temples,	• Collect data related to the implementation of the action plan and provide feedback to key agencies/departments. • Participate in an annual evaluation of heat wave action plan. Review the revised heat wave action plan.

		• Urban forestry, avenue plantation, and encouraging roof gardens to increase the green cover in urban areas to reduce heat levels. • Special care in restricting outdoor activities and functions during the heat period. • Identification of NGOs / Rotary Clubs / Lions Clubs and Corporate houses (under Corporate Social Responsibility) to provide shelters, drinking water, medical supplies, and temporary homes during heat days. • Explore mitigation measures, like cool roofing, to be implemented in urban areas.	schools, and State Government or Local body, run temporary night shelters for those without house or access to water and/or electricity at home. • Expand access to shaded areas for outdoor workers, slum communities, and other vulnerable sections of the population. • Keep open the parks for a longer duration during evenings. • All non-essential uses of water (other than drinking, and keeping cool) may be suspended, if necessary. • Distribution of fresh drinking water to the public by opening water centers (Drinking water shelters) at people's congregation points like marketplaces, labouradda, Bus stations, etc. Water may be	
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			distributed through pouches to the poor in the identified high-risk areas. • Actively involve NGOs, lions club, rotary club, and corporate houses in providing shelter and drinking water facilities.	
5.	Labour Department	• Organize training for employers, outdoor laborers and workers on the health impacts of extreme heat and protective measures to be taken during high-temperature periods. • Utilize maps of construction sites and outdoor work spots preferably overlaying with irradiation maps from IMD or heat island maps to identify more high-risk outdoor workers and to conduct publicity campaigns during high-risk days. • Preparing a list of factory medical officers, contractors, and house-side	• Encourage employers to shift outdoor workers' schedules away from peak afternoon hours (12 – 4 pm) during a heat alert. • To ensure emergency ice packs and heat-illness prevention materials to construction workers. • Ensure provision of shelters/ cooling areas, water, and supply of emergency medicines like ORS, IV fluids, etc. at work sites by employers.	• Collect data related to the implementation of the action plan and provide feedback to key agencies/departments. • Participate in an annual evaluation of heat wave action plan. Review the revised heat wave action plan.

		non-factory workers to include in heat alert and active communication. • Heat illness orientation planning for factory medical officers.		
6.	Rural Development and Panchayat	• Collecting information on the works sanctioned under the MGNREGS program and other schemes in High-risk areas to plan for mitigation efforts during the heat period. • To ensure shelter and supply of adequate drinking water at work spots.	• Reschedule working hours to avoid intense heat timings in all the works sanctioned under MGNREGS. • Provision of water and shelters/cooling areas wherever necessary.	• Collect data related to the implementation of the action plan and provide feedback to key agencies/departments. • Participate in an annual evaluation of heat wave action plan. Review the revised heat wave action plan.
7.	Animal Husbandry, Dairy Development and Fisheries	• Preparation of posters & pamphlets with tips to take care of cattle and poultry during heat waves. • Publicity of protective measures to save cattle and poultry during heat periods through District heads and Farmer Training Centres.	• Display posters and distribute pamphlets on the precautionary measures to be taken to safeguard cattle and poultry birds during heat periods in villages and important junctions. • Ensure adequate stock of medicines in all veterinary hospitals.	• Collect data related to the implementation of the action plan and provide feedback to key agencies/departments. • Participate in an annual evaluation of heat wave action plan. Review the

		• Checking inventory of necessary medicines for the treatment of cattle and poultry. • Preparation of plans to provide drinking water for cattle in case of scarcity	• Ensure the visit of field staff during heat waves to villages for follow-up action in the treatment of cattle/poultry birds.	revised heat wave action plan.
8.	Transport	• Obtaining lists of risk areas and review of Bus timings and available shelters in the high-risk areas. • Planning for shade/shelter, drinking water, and fans in the waiting areas of passengers. • Review plan with cab operator/auto/transport associations and also highway patrol • Display of precautionary measures (Do's and don'ts) on busses, autos, in bus stations & auto stands and distribution of pamphlets to passengers. • Planning to provide ORS, Ice packets, etc., and medical services in Bus stations.	• Display posters & distribute pamphlets on the prevention of heat-related illness in bus stands, auto stands, etc. • Ensure availability of shade/shelters, drinking water, ORS packets, etc., in bus stands, auto stands, etc. • Establish health teams at major bus stands / Terminals and other public places to ensure the availability of water and ORS packets in long-distance buses. • Do not run buses as per as possible during peak hours (12-4 pm) when a heat wave is declared.	• Collect data related to the implementation of the action plan and provide feedback to key agencies/departments. • Participate in an annual evaluation of heat wave action plan. Review the revised heat wave action plan.

9.	School Education	• Designing child-friendly educational preventative training and distributing heat protection materials at local schools. • Training of schoolteachers to equip them with knowledge of heat protection tips and activities which they can disseminate in classrooms. • Scheduling of examinations before starting of normal Heat period.	• Ensure supply of water for students and teachers if school is functioning. • If school is not functioning, permit use of school premises as shelter during daytime.	• Collect data related to the implementation of the action plan and provide feedback to key agencies/departments. • Participate in an annual evaluation of heat wave action plan. Review the revised heat wave action plan.
10.	Social Security and Development of Women and Children	• Setting up of nutritional resource centres at Anganwadi centres to supplement nutritional deficiency in children. • Pre heat wave necessary precautionary methods such as provision of proper stock of ORS, buttermilk and other rehydration methods may be arranged well in advance as the heatwave extends for about 17-45 days.	• Display IEC materials at Anganwadis and encourage Integrated Child Development Scheme (ICDS) workers to disseminate Heat Wave related information with special focus on infants, children below five years, pregnant and lactating mothers, and geriatric population to protect them from dehydration.	• Collect data related to the implementation of the action plan and provide feedback to key agencies/departments. • Participate in an annual evaluation of heat wave action plan. Review the revised heat wave action plan.

		• To create a surveillance mechanism on tracking children, lactating mothers and women through ICDS and Anganwadi centres in the state. • Capacity building of Anganwadi sevikas, Asha workers, ANM nurses and ICDS workers to identify symptoms in women and children and to report it when necessary. • To identify the districts or villages where high child mortality rates are present to take necessary precautionary methods.	• Provision of drinking water and first aid at all the Anganwadi Centers, old age homes, orphanages.	
11.	Tourism and Cultural Affairs	• Ensure proper registration of tourists who are visiting the State during heat season. • Ensure availability of heat relief measures at tourist places. • Display of Heat Wave precautionary measures for tourists during summer at	• Ensure the availability of drinking water and cool resting sheds. • Restrict the timing of the visit of tourist places during peak summer days.	• Collect data related to the implementation of the action plan and provide feedback to key agencies/departments. • Participate in an annual evaluation of heat wave action plan. Review the

		tourist points and related information in website of tourism. • Provision of funds for Heat Wave management.		revised heat wave action plan.
12.	Local Government / Fire Services	• To check the readiness of vehicles and firefighting equipment to face any fire emergency.	• Ensure the presence of staff during the heat alert period, if necessary, by restricting leaves. • Ensure the functioning of communication equipment to receive messages/alerts of the occurrence of fire. • Ensure an adequate supply of water and foam to fight fires.	• Collect data related to the implementation of the action plan and provide feedback to key agencies/departments. • Participate in an annual evaluation of heat wave action plan. Review the revised heat wave action plan.
13.	Community groups / Self-help groups/ Ward-level committees / NGOs	• Conduct training programs, workshops, and outreach sessions with community / Self-help groups and mobilize ASHA workers, Anganwadis, and Ward Committees in Municipalities to help inform and get vulnerable communities more actively involved in mitigating Heatwave consequences.	• Take all precautions to avoid heat-related illness. • Keep cool and hydrated during the heat season by drinking water, staying out of the sun, and wearing light clothing. • Check on vulnerable neighbours, particularly during a heat alert.	• Reach the unreached and educate the community on a continuous basis, in addition to providing feedback on the outreach and impact of heat wave action plan to the key departments / agencies /

		• Identification of NGOs, Voluntary Organizations in reaching out to the Public, especially vulnerable groups. • Encourage discussions for finding early signs of heat exhaustion with local doctors or Health Centre. • Inform fellow community members about how to keep cool and protect oneself from heat.	• Limit heavy work in direct sun or indoors, if poorly ventilated, especially during a heat alert.	nodal officers at State and District Levels.
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Annexure 6: Heat Illness - Treatment Protocol

Recognizing that treatment protocols may vary slightly depending on the specific healthcare setting (such as EMS, health center, clinic, hospital emergency department, etc.), the following guidelines should generally apply to all patients with heat-related illnesses:

- 1) Conduct an initial patient assessment primary survey, which includes evaluating the airway, breathing, circulation, disability, and exposure. Measure vital signs, including temperature.
- 2) Consider the possibility of heat illness in the differential diagnosis if:
 - a) The patient presents with suggestive symptoms and signs.
 - b) The patient exhibits one or more of the following risk factors:
 - i) Extremes of age (infants, elderly).
 - ii) Debilitation, physical deconditioning, overweight or obesity.
 - iii) Lack of acclimatization to environmental heat (recent arrival or early in the summer season).
 - iv) Any significant underlying chronic disease, including psychiatric, cardiovascular, neurologic, hematologic, obesity, pulmonary, renal, and respiratory disease.
 - v) The patient is taking one or more of the following: - Sympathomimetic drugs - Anticholinergic drugs - Barbiturates - Diuretics - Alcohol
- 3) Remove the patient from environmental heat exposure and discontinue physical activity. Initiate passive cooling procedures:
 - a) Apply cool wet towels or ice packs to the axillae, groin, and around the neck. If the patient's condition is stable, a cool shower may be considered but evaluate the potential risks against the benefits and availability of other cooling measures.
 - b) Spray cool water or gently pat cool water onto the skin. iii. Use a fan to blow cool air onto moist skin.
- 4) If the patient's temperature is below 40°C, repeat the assessment every 5 minutes. If the patient's condition is improving, attempt oral hydration with clear liquids. Oral rehydration solutions (ORS) can be used but are not necessary. Cool liquids are preferable to cold liquids. Observe the patient closely.
- 5) If the patient's temperature is 40°C or above, initiate intravenous (IV) rehydration and immediately transport the patient to the emergency department for further stabilization.

Annexure 7: NPCCHH Public Health Advisory for Extreme Heatwave

DO's

For general population

1. Stay hydrated:
 - Drink sufficient water whenever possible, even if you are not thirsty. Thirst is not a good indicator of dehydration.
 - Carry drinking water when traveling.
 - Use Oral Rehydration Solution (ORS), and consume homemade drinks like lemon water, butter milk/lassi, fruit juices with some added salt.
 - Eat seasonal fruits and vegetables with high water content like watermelon, musk melon, orange, grapes, pineapple, cucumber, lettuce or other locally available fruits and vegetables.
2. Stay covered:
 - Wear thin, loose, cotton garments, preferably light-coloured.
 - Cover your head: use an umbrella, hat, cap, towel, and other traditional head gear during exposure to direct sunlight.
 - Wear shoes or chappals while going out in the sun.
3. Stay alert.
 - Listen to the Radio; watch TV; read the Newspaper for local weather news. Get the latest update on weather on the India Meteorological Department (IMD) website at <https://mausam.imd.gov.in/>
4. Stay indoors/in the shade as much as possible:
 - In well-ventilated and cool places
 - Block direct sunlight and heat waves: Keep windows and curtains closed during the day, especially on the sunny side of your house. Open them up at night to let cooler air in.
 - If going outdoors, limit your outdoor activity to cooler times of the day, i.e., morning and evening.
 - Reschedule or plan outdoor activities during cooler parts of the day.

For vulnerable population

1. Although anyone at any time can suffer from heat stress & heat-related illness, some people are at greater risk than others and should be given additional attention. These include:
 - a. Infants and young children
 - b. People who have a mental illness
 - c. Physically ill, especially with heart disease or high blood pressure
 - d. People working outdoors
 - e. Pregnant women
2. Travelers from colder climates to hot climates should allow one week's time for their bodies to acclimatize to heat, avoid overexertion, and drink plenty of water. Acclimatization is achieved by a gradual increase (over 10-15 days) in exposure/physical activity in a hot environment.
3. Elderly or sick people living alone should be supervised and their health monitored on a daily basis.

Other precautions

1. Keep your home cool; use curtains, shutters, or sunshades and open windows at night.
2. Try to remain on lower floors during the day.
3. Use a fan, spray bottles, damp cloths, and ice towels to cool down the body.
4. Immersing feet in 20°C water above the ankle provides rapid cooling by reducing dehydration and thermal discomfort.

Don'ts

1. Avoid getting out in the sun, especially between 12:00 noon and 03:00 pm.
2. Avoid strenuous activities when outside in the afternoon.
3. Do not go out barefoot.
4. Avoid cooking during peak summer hours. Open doors and windows to ventilate the cooking area adequately.
5. Avoid alcohol, tea, coffee, and carbonated soft drinks or drinks with large amounts of sugar- as these lead to loss of more body fluid or may cause stomach cramps.
6. Avoid high-protein food and do not eat stale food.
7. Do not leave children or pets in parked vehicles. The temperature inside a vehicle could get dangerous

Annexure 8: Details of Deaths reported due to Heat Wave (record kept with State Government)

S. No.	Name and Address	Age	Sex (M/F)	Occupation	Place of Death	Date and Time of Death	Max Temp Recorded (Rectal and Oral)	Death reported during Heat wave Period (Y/N)	List of chronic diseases (Ask family members)	Date and time of Post Mortem (if conducted)	Date and time of joint enquiry conducted with a revenue authority	Cause of Death	Remarks
1													
2													
3													
4													

Annexure 9: Daily Report of Heat Stroke Cases and Deaths (District report to State Government)

S.No.	Village	PHC	Block/City	Name & Son/ Daughter/Wife of	Urban/ Rural (U/R)	BPL (Y/N)	Age/Sex	Date of attack of Heat Stroke	Any Antecedent illness	Cause of death	Death confirmed by MOs and MROs

उष्णतेच्या लाटेच्या प्रभावास प्रतिसाद देण्यासाठी मार्गदर्शक सूचना

राज्यातील विविध जिल्ह्यात दरवर्षी उष्णतेच्या लाटेचा प्रभाव जाणवतो. उष्णतेच्या लाटांमुळे आरोग्यविषयी गंभीर समस्या निर्माण होण्याची संभावना लक्षात घेता, जिल्हा स्तरावरून उष्णतेच्या लाटेचा प्रभाव कमी करण्यासाठी संबंधित प्रशासकीय विभाग, महानगरपालिका/ नगरपालिका व जिल्हा आपत्ती व्यवस्थापन प्राधिकरणांनी करावयाच्या योजनासाठी खालील मार्गदर्शक सूचनांचा वापर करावा.

१. महानगरपालिका/परिषद

- बाजार, प्रमुख कार्यालये, बस स्टँड, टॅक्सी स्टँड, रिक्षा स्टँड इत्यादी सार्वजनिक ठिकाणी पुरेशी सावलीची व्यवस्था करावी तसेच सदर ठिकाणी पिण्याच्या पाण्याची सोय करावी (पंखा/कुलर इ. नादुरुस्त असल्यास दुरुस्त करावे)
- सार्वजनिक ठिकाणी उष्णतेच्या लाटेशी संबंधित घ्यावयाची काळजी, संदर्भात पोस्टर्स/बॅनर्स लावावेत.
- उष्माघातापासून बचाव करण्यासाठी सार्वजनिक ठिकाणी प्रथमोपचार पेठ्या ठेवाव्यात आणि त्याच्या वापराबाबत आवश्यक सूचना लिहाव्यात.
- टॅक्सी स्टँड, बस स्टँड (सरकारी/खाजगी), बाजार क्षेत्र आणि सरकारी कार्यालये या सार्वजनिक ठिकाणी पिण्याचे थंड पाणी उपलब्ध करावे.
- महानगरपालिका आयुक्त आणि मुख्य कार्यकारी अधिकारी यांनी शॉपिंग मॉल्स मध्ये ग्राहकांसाठी आणि बाहेरील लोकांसाठी थंड पिण्याच्या पाण्याची सुविधा करावी. तसेच बाहेरील लोकांना पाण्याच्या उपलब्धतेबद्दल माहिती देण्यासाठी मार्गदर्शक फलक ठेवावे.
- उपलब्ध असल्यास रस्त्यावर पाणी शिंपडावे.
- सर्व उद्याने दुपारी १२:०० ते दुपारी ४: ०० वेळेत खुली ठेवावीत.

२. आरोग्य विभाग

- उष्णतेच्या लाटेशी संबंधित आजार निरीक्षण आणि अहवाल तयार करणेसाठी नोडल अधिकारी नियुक्त करावे.
- प्रत्येक आरोग्य केंद्राच्या परिसरात उष्णतेच्या लाटेपासून बचाव करण्या संबंधित माहिती फलक लावावेत.

- उष्माघाताच्या रुग्णांसाठी आरोग्य केंद्रांमध्ये स्वतंत्र वैद्यकीय वॉर्डाची व्यवस्था करावी.
- उष्माघाताच्या रुग्णांच्या उपचारासाठी जलद प्रतिसाद टीम (RRT) तयार करावी. बहुउद्देशीय कामगार, आशा वर्कर्स यांना उष्माघाताच्या रुग्णांचे उपचार आणि तपशील गोळा करण्याच्या सूचना देण्यात याव्यात. उपलब्ध माहिती तालुका आरोग्य अधिकारी यांच्याकडे पाठवावी.
- रुग्णवाहिका सार्वजनिक ठिकाणी, विशेषतः दुपारी तत्पर ठेवावी.
- प्राथमिक आरोग्य केंद्र (PHCs) आणि इतर स्थानिक रुग्णालयांमध्ये ORS पावडरसह आवश्यक वैद्यकीय सुविधा यांचा साठा पुरेसा ठेवावा.
- लहान मुले, अपंग व्यक्ती, स्त्रिया, वृद्ध लोक आणि मजूर यांसारख्या असुरक्षित गटांसाठी विशेष काळजी घेण्याची व्यवस्था करावी.
- उष्माघाताशी संबंधित प्रकरणे आणि मृत्यूचे दैनिक अहवाल करावे.
- ज्या कार्यक्रमांमध्ये जेवण दिले जाणार आहे, अशा कार्यक्रमांच्या आयोजकांसाठी आरोग्य विभागाचे ना हरकत प्रमाणपत्र अनिवार्य करावी.

३. पंचायत विभाग

- पंचायत स्तरावर जागरूकता आणि उष्णतेच्या लाटेपासून बचाव करण्याच्या उपायांचा प्रसार करावा.
- मनरेगा कामगारांच्या कामाचे तासांचे नियोजन करा आणि कामाच्या ठिकाणी सावली आणि थंड पाणी द्यावे.
- उष्णतेच्या सतर्कतेच्या वेळी निवार्याची सोय करावी आणि पिण्याच्या पाण्याची उपलब्धता ठेवावी.
- आरोग्य विभागाशी समन्वय साधा आणि ब्लॉक स्तरावर प्रशिक्षण सत्र आयोजित करावे.
- उष्णतेच्या लाटेबाबत जनजागृतीसाठी ग्रामसभेचा प्रभावी माध्यम म्हणून वापर करण्यात यावा.

४. शिक्षण विभाग (प्राथमिक आणि माध्यमिक)

- हवामान खात्याकडून आलेल्या चेतावणीनुसार शाळा/महाविद्यालयांचा वेळा नियोजन करा आणि थंड वर्गखोल्या, प्रथमोपचार आणि पिण्याच्या पाण्याची सोय करावी.
- शाळेच्या वेळांमध्ये बदल करून उष्णतेच्या लाटेच्या परिस्थितीनुसार सुट्टी देण्यात याव्या.
- उन्हाळ्यात विद्यार्थ्यांनी मैदानी/ शारीरिक हालचाली टाळाव्यात. बाहेर मैदानात वर्ग घेऊ नयेत.

- दुपारच्या सत्रात मैदानावरील खेळांचे नियोजन करू नका.
- विद्यार्थ्यांना उष्णतेच्या लाटेपासून बचाव करण्यासाठी शिक्षित करा आणि आपत्कालीन सेवांचा संपर्क द्या.
- परीक्षा सकाळच्या सत्रातच घेतल्या जाव्यात.
- पंखे सुस्थितीत राहतील याची खात्री करावी.
- माध्यान्ह भोजन योजनेअंतर्गत मुलांना सरबत, ताक आणि ORS पॅकेट द्यावे.

५. कामगार विभाग

- कामगारांवर उष्णतेच्या लहरींचा प्रभाव कमी करण्यासाठी वेळोवेळी सूचना द्या आणि आवश्यक सुविधा द्याव्या.
- कामगारांचे कामाच्या तासांचे नियोजन करा आणि कामाच्या ठिकाणी सावली आणि थंड पाणी उपलब्ध करून द्यावा.
- कामगारांना उष्माघात प्रथमोपचार आणि CPR प्रशिक्षण द्यावे.

६. परिवहन विभाग

- कमाल उष्णतेच्या लाटे दरम्यान सार्वजनिक वाहतुकीच्या सेवा सुरळीत राहिल आणि प्रतीक्षा क्षेत्रावर सावली, पाणी आणि पंखे याची सोय करावी.
- वाहनांवर खबरदारीचे उपाय दाखवा आणि प्रवाशांना माहितीपत्रक वितरित करावे.
- दुरुचित्रफीत, IEC प्रसारित करण्यासाठी बसस्थानके इत्यादींचा सार्वजनिक संबोधन प्रणालीचा वापर करावा.
- उन्हाळ्यात सर्व बसेसमध्ये प्रथमोपचार किट (ORS च्या मुबलक साठ्यासह) असल्याची खात्री करावी.
- बस स्थानकावर पाण्याची सोय, पंखे सुस्थितीत असावे.

७. वन विभाग

- वन्य प्राणी आणि पक्ष्यांसाठी पुरेसे वृक्षारोपण आणि पाण्याचा पुरवठा असावा.
- आग रोखण्यासाठी वनक्षेत्रांचे निरीक्षण करा आणि आवश्यक प्रतिबंधात्मक उपाय करावे.

८. पाणी पुरवठा व स्वच्छता विभाग

- ग्रामीण भागात पिण्याच्या पाण्याची सुविधा करावी आणि बोअरहोल आणि तलावांची तातडीने दुरुस्ती करावी.
- उन्हाळी हंगामापूर्वी सर्व प्रकारच्या जलस्रोतांचे तपासणी आणि देखभाल करावी.
- सकाळी किंवा संध्याकाळी पाणी पुरवठ्याच्या वेळेचे नियोजन करावे. या सूचना विशेषतः सार्वजनिक पाण्याच्या वापरा संदर्भात ग्राम पंचायत स्तरावर देण्यात याव्यात.

९. सार्वजनिक बांधकाम विभाग

- सरकारी इमारती आणि संस्थांमध्ये कूल रूफ पेंटच्या वापराबद्दल प्रचार करावा.
- बांधकामाच्या ठिकाणी मजुरांसाठी निवारा आणि प्रथमोपचार किट उपलब्ध करावे.

१०. ऊर्जा विभाग

- उष्णतेच्या लाटांमध्ये वीजपुरवठा अखंडित राहावा यासाठी सर्व उपाय योजना करण्यात याव्या.
- वीज बिलांवर उष्णता लहरीशी संबंधित IEC सामग्री प्रसारित करावी किंवा वीज बिलांसह IEC-संबंधित माहितीपत्रक वितरित करावे.

११. कृषी विभाग

- उष्णतेमुळे पिकांचे नुकसान कमी करण्यासाठी किंवा टाळण्यासाठी शेतकऱ्यांना मार्गदर्शन करावे.
- शेतकऱ्यांपर्यंत हवामानाची माहिती प्रसारित करण्यासाठी "शेतकरी मित्र" यांचा वापर करावा.
- शेत तलावांच्या देखभालीसाठी किंवा निर्मितीला प्रोत्साहन द्यावे. शेतीवर आधारित उपजीविकेसाठी पाण्याची उपलब्धते संदर्भात नियोजन करावे.

१२. पोलीस विभाग

- पोलिसांसाठी सिग्नलवर निवारा किंवा बूथ उभारणी करावी.
- पोलीस स्टेशन आणि ट्रॅफिक बूथवर प्रथमोपचार किट उपलब्ध करावे.
- प्रत्येक पोलीस स्टेशनवर एका पोलीस कर्मचार्यासाठी उष्माघात प्रथमोपचार आणि CPR प्रशिक्षण आयोजित करावे.
- दुपारी आयोजित केलेल्या ओपन टू स्काय कार्यक्रमांना परवानगी पत्र देऊ नका.
- कोणत्याही सार्वजनिक कार्यक्रमात मुबलक पाण्याची व्यवस्था तसेच जागेच्या उपलब्धते प्रमाणे कार्यक्रमांना परवानगी द्यावी.

- हवामान खात्याकडून आलेल्या चेतावणीनुसार उष्मालाट असलेल्या दिवशी मोठ्या कार्यक्रमांना परवानगी देऊ नये.

१३. महिला व बाल विकास विभाग

- आशा वर्कर्स आणि अंगणवाडी सेविका यांच्यासाठी उष्माघात प्रथमोपचार आणि CPR प्रशिक्षण आयोजित करावी.
- ICDS योजनेंतर्गत, सरबत, ताक आणि ORS पॅकेट मुलांना आणि गरोदर आणि स्तनदा महिलांना द्यावे.
- ग्रामपंचायत स्तरावर उष्णतेच्या लाटेबाबत आशा कार्यकर्त्या आणि अंगणवाडी सेविकांद्वारे द्वारे जागरूकता निर्माण करावी.

१४. महाराष्ट्र औद्योगिक विकास महामंडळ (MIDC)

- सर्व उद्योगांनी मजुरांना पुरेसा निवारा आणि थंड पिण्याच्या पाण्याची सुविधा उपलब्ध करून द्यावी.
- सर्व सुरक्षा अधिकाऱ्यांना उष्माघात प्रथमोपचार आणि CPR प्रशिक्षण द्या. उष्णतेच्या लाटेपासून बचाव संबंधित फलक प्रत्येक उद्योगाच्या बाहेर लावावेत.
- उद्योगाच्या नियोजन आरखड्यात आपत्कालीन योजनेमध्ये उष्मा लहरी सज्जतेचा अध्याय समाविष्ट करावा.
- औद्योगिक क्षेत्रात वेळोवेळी आगीच्या घटना घडतात त्यामुळे पुरेशा सुविधा पाण्याचा साठा तसेच फोम टेंडर ची सुविधा ठेवावी.

१५. पशुधन विभाग

- उष्णतेच्या लाटेत स्थानिक लोकांना प्राण्यांची काळजी घेण्यासाठी फलक आणि माहितीपत्रक तयार करावी.
- प्राणी संरक्षणासाठी जागरूकता कार्यक्रम आयोजित करा आणि पशुवैद्यकीय औषधांचा पुरेसा साठा ठेवावा आणि जनावरांसाठी पाण्याची व निवार्याची सोय करावी.
- मोठ्या संख्येने पशुधन असलेल्या क्षेत्रात हिरवा चारा आणि थंड पाण्याची उपलब्धता करावी.

१६. जिल्हा आपत्ती व्यवस्थापन प्राधिकरण

- जिल्हा आपत्ती व्यवस्थापन प्राधिकरणाने एक सर्वसमावेशक “उष्णतेची लाट कृती आराखडा”, व मानक कार्यप्रणाली (SoPs) आणि घटना प्रतिसाद प्रणाली (IRS) विकसित करावी. ज्यामध्ये लोकांच्या जीवनावर आणि उपजीविकेवर उष्णतेच्या लाटेचा प्रभाव कमी करण्यासाठी विविध जिल्हा भागधारकांच्या भूमिका आणि जबाबदाऱ्या निश्चित कराव्यात.
- जिल्हा आपत्ती व्यवस्थापन प्राधिकरणाने उष्णतेच्या लाटेच्या संदर्भात जिल्हा स्तरावर उष्मालाट नोडल अधिकारी नियुक्त करावा.
- वृद्ध लोकसंख्येसाठी छत्री, पादत्राणे आणि कापसाचे टॉवेल स्वेच्छेने देणगी देण्यासाठी एनजीओ, सीएसओ आणि सीएसआरकडे संपर्क साधावा.
- सामाजिक माध्यमातून IEC साहित्य प्रसारित करण्यासाठी जिल्हा माहिती कार्यालयाचा वापर करावा.

अति उष्णतेच्या लाटेसाठी हवामान बदल आणि मानवी आरोग्यावरील राष्ट्रीय कार्यक्रम
(NPCCHH) यांचा सार्वजनिक आरोग्य सल्ला

सामान्य लोकांसाठी

‘हे’ करा

- जेव्हा शक्य असेल तेव्हा पुरेसे पाणी प्या, तुम्हाला तहान लागली नसली तरीही. तहान हे निर्जलीकरणाचे चांगले सूचक नाही.
- प्रवास करताना पिण्याचे पाणी सोबत ठेवावे.
- ओरल रीहायड्रेशन सोल्यूशन (ORS) वापरा आणि लिंबू पाणी, ताक/लस्सी आणि फळांचे रस यांसारखे घरगुती पेये प्यावे.
- टरबूज, खरबूज, संत्री, द्राक्षे, अननस, काकडी, कोशिंबिर तसेच स्थानिक उपलब्ध फळे आणि भाज्या यासारखी उच्च पाणी सामग्री असलेली हंगामी फळे आणि भाज्या खाव्या.
- पातळ, सैल, सुती आणि शक्यतो हलक्या रंगाचे कपडे घालावे.
- तुमचे डोके झाकून ठेवा: थेट सूर्यप्रकाशाच्या संपर्कात असताना छत्री, टोपी, टॉवेल आणि इतर पारंपारिक साधनांचा वापरावे.
- उन्हात बाहेर जाताना शूज किंवा चप्पल घालाव्यात.
- रेडिओ, टीव्ही वर प्रसारित होणाऱ्या उष्णतेच्या लाटेबाबत बचाव उपायावर लक्ष द्या आणि स्थानिक हवामान बातम्यांसाठी वर्तमानपत्र वाचावीत.
- भारतीय हवामान विभागाच्या (IMD) वेबसाइट <https://mausam.imd.gov.in/> वर हवामानाविषयी अद्ययावत माहिती घेत रहावे.
- हवेशीर आणि थंड ठिकाणी जास्त वेळ घालवा.
- थेट सूर्यप्रकाश आणि उष्णतेच्या लाटा रोखा: – दिवसा खिडक्या आणि पडदे बंद ठेवावे, विशेषतः तुमच्या घराच्या सूर्यप्रकाशाच्या बाजूला. थंड हवा येण्यासाठी त्यांना रात्री खिडक्या उघड्या कराव्यात.
- जर तुम्ही घराबाहेर जात असाल, तर तुमची घराबाहेरील कामे दिवसाच्या सकाळ आणि संध्याकाळ वेळेपर्यंत मर्यादित करावी.

- दिवसाच्या थंड वेळांमध्ये बाहेरील कामांचे नियोजन करावे.

संवेदनशील किंवा जास्त जोखीम असलेल्या लोकांसाठी

- जरी कोणालाही उष्णतेचा ताण आणि उष्णतेशी संबंधित आजाराचा त्रास होऊ शकतो, तरीही काही लोकांना इतरांपेक्षा जास्त धोका असतो आणि त्यांच्याकडे अतिरिक्त लक्ष द्यावे.
- समाविष्ट घटक
 - ❖ लहान अर्भक आणि लहान मुले
 - ❖ घराबाहेर काम करणारे लोक
 - ❖ गर्भवती महिला
 - ❖ ज्यांना मानसिक आजारपण असेलेल व्यक्ती
- शारीरिकदृष्ट्या आजारी, विशेषतः हृदयरोग किंवा उच्च रक्तदाब
- थंड हवामानापासून ते उष्ण हवामानापर्यंतच्या प्रवाश्यांनी त्यांच्या शरीराला उष्णतेला अनुकूल होण्यासाठी एक आठवडा वेळ द्यावा, जास्त श्रम टाळावेत आणि भरपूर पाणी प्यावे.

इतर खबरदारी

- एकटे राहणाऱ्या वृद्ध किंवा आजारी व्यक्तींवर देखरेख ठेवली पाहिजे आणि त्यांच्या आरोग्याचे दररोज निरीक्षण केले पाहिजे.
- तुमचे घर थंड ठेवा; पडदे, शटर किंवा सनशेड वापरा आणि रात्री खिडक्या उघडा.
- दिवसा खालच्या मजल्यावर राहण्याचा प्रयत्न करावा.
- शरीर थंड करण्यासाठी पंखा, स्प्रे बाटल्या, ओलसर कापड आणि बर्फाचे टॉवेल वापरावा.
- पाण्यात पाय बुडवल्याने निर्जलीकरण आणि शरीराची उष्णतेमुळे होणारी अस्वस्थता कमी होऊन जलद थंडावा मिळतो.

‘हे’ करू नका

- उन्हात बाहेर पडणे टाळा, विशेषतः दुपारी १२:०० ते ०३:००.
- दुपारच्या वेळी बाहेर असताना कठोर परिश्रम टाळा.
- अनवाणी बाहेर जाऊ नका.

- अति उष्णवेळेत स्वयंपाक करणे टाळा. स्वयंपाक क्षेत्र पुरेशा प्रमाणात हवा येणासाठी दरवाजे आणि खिडक्या उघड्या ठेवाव्यात.
- अल्कोहोल, चहा, कॉफी आणि कार्बोनेटेड सॉफ्ट ड्रिंक्स किंवा मोठ्या प्रमाणात साखर असलेले पेय टाळा- कारण यामुळे शरीरातील जास्त पाणी कमी होते किंवा पोटात पेटके येऊ शकतात.
- उच्च प्रथिनेयुक्त अन्न टाळा आणि शिळे अन्न खाऊ नका.
- उभ्या केलेल्या वाहनांमध्ये लहान मुले किंवा पाळीव प्राणी सोडू नका. वाहनातील तापमान धोकादायक ठरू शकते.

Public Health Advisory: Extreme Heat/Heatwave

Do's

For general population

Stay hydrated:

- Drink sufficient water whenever possible, even if you are not thirsty. Thirst is not a good indicator of dehydration.
- Carry drinking water when traveling
- Use Oral Rehydration Solution (ORS), and consume homemade drinks like lemon water, butter milk/lassi, fruit juices with some added salt.
- Eat seasonal fruits and vegetables with high water content like water melon, musk melon, orange, grapes, pineapple, cucumber, lettuce or other locally available fruits and vegetables.

Stay covered:

- Wear thin loose, cotton garments preferably light coloured
- Cover your head: use umbrella, hat, cap, towel and other traditional head gears during exposure to direct sunlight
- Wear shoes or chappals while going out in sun

Stay alert:

- Listen to Radio; watch TV; read Newspaper for local weather news. Get the latest update of weather on India Meteorological Department (IMD) website at <https://mausam.imd.gov.in/>

Stay indoors/in shade as much as possible:

- In well ventilated and cool places
- Block direct sunlight and heat waves: Keep windows and curtains closed during the day, especially on the sunny side of your house. Open them up at night to let cooler air in.
- If going outdoor, limit your outdoor activity to cooler times of the day i.e., morning and evening
- Reschedule or plan outdoor activities during cooler parts of the day.

For vulnerable population

Although anyone at any time can suffer from the heat stress & heat-related illness, some people are at greater risk than others and should be given additional attention.

These include:

- Infants and young children
- Pregnant women
- Physically ill, especially with heart disease or high blood pressure
- Travelers from colder climate to a hot climate should allow one week's time for their bodies to acclimatized to heat, avoid overexertion, and should drink plenty of water. Acclimatization is achieved by gradual increase (over 10-15days) in exposure/physical activity in hot environment
- People working outdoors
- People who have a mental illness

Other precautions

- Elderly or sick people living alone should be supervised and their health monitored on a daily basis.
- Keep your home cool, use curtains, shutters or sunshade and open windows at night.
- Try to remain on lower floors during the day.
- Use fan, spray bottles, damp cloths, ice towels to cool down body.
- Immersing feet in 20°C water above ankle provides rapid cooling by reducing dehydration and thermal discomfort.

Don'ts

- Avoid getting out in the sun, especially between 12:00 noon and 03:00 pm
- Avoid strenuous activities when outside in the afternoon
- Do not go out barefoot
- Avoid cooking during peak summer hours. Open doors and windows to ventilate cooking area adequately
- Avoid alcohol, tea, coffee and carbonated soft drinks or drinks with large amount of sugar- as these actually, lead to loss of more body fluid or may cause stomach cramps
- Avoid high-protein food and do not eat stale food
- Do not leave children or pets in parked vehicle. Temperature inside a vehicle could get dangerous.

For Employers and workers

- Provide cool drinking water at work place and remind them to drink a cup of water every 20 minutes or more frequently to stay hydrated
- Caution workers to avoid direct sunlight
- Provide shaded work area for workers. Temporary shelter can be created at work site.
- Schedule strenuous and outdoor jobs to cooler times of the day i.e., morning and evening hours
- Increase the frequency and length of rest breaks for outdoor activities- at least every 5 minutes after 1 hour of labour work
- Listen to Radio; watch TV; read Newspaper for local weather news and act accordingly. Get the latest update of weather on India Meteorological Department (IMD) website at <https://mausam.imd.gov.in/>
- Assign additional workers or slow down the pace of work
- Make sure everyone is properly acclimatized: it takes weeks to acclimatize to a hotter climate. Do not work for more than three hours in one day for the first five days of work. Gradually increase the amount and time of work.
- Train workers to recognize factors which may increase the risk of developing a heat related illness and the signs and symptoms of heat stress and start a "buddy system" since people are not likely to notice their own symptoms
- Trained First Aid providers should be available and an emergency response plan should be in place in the event of a heat-related illness.
- Pregnant workers and workers with a medical condition or those taking certain medications should discuss with their physicians about working in the heat.
- If working outdoors wear light-coloured clothing preferably long sleeve shirt and pants, and cover the head to prevent exposure to direct sunlight.
- Organize awareness campaigns for employees
- Install temperature and forecast display at the workplace.
- Distribute informational pamphlets and organize training for employers and workers regarding health impacts of extreme heat and recommendations to protect themselves during high temperatures.

Precautions During Mass gathering/Sport event

- Outdoor/indoor crowded situations increase risk of acute heat-related illnesses (HRI) even in absence of active heatwave alerts in the area.
- Physical exertion, direct sun exposure, overcrowding, and difficult access to water, food and shade may worsen health in vulnerable groups.
- Attendees should stay hydrated, cool, be aware of HRI signs, symptoms and seek medical care.

Health Impact of Heat: Heat-Related Illnesses

- Normal human body temperature ranges between 36.4°C to 37.2°C (97.5°F to 98.9°F)
- Exposure to high outdoor and/ indoor temperatures can induce heat stress, directly and indirectly, leading to heat-related illnesses
- Heat-related illnesses include (from mild to severe)—**heat rash** (prickly heat), **heat oedema** (swelling of hands, feet and ankles), **heat cramps** (muscle cramps), **heat tetany**, **heat syncope** (fainting), **heat exhaustion**, and **heat stroke**.
- Heat stress may also exacerbate chronic diseases like cardiovascular, respiratory, kidney diseases

Watch out for symptoms of heat stress, which include:

- dizziness or fainting;
- nausea or vomiting;
- headache
- extreme thirst
- decreased urination with unusually dark yellow urine
- rapid breathing and heartbeat

Heat-related illnesses are preventable

If you or others feel unwell and experience any of above symptoms during extreme heat,

- Immediately move to a cool place and drink liquids. **Water is best.**
- Get help/medical attention
- Measure your body temperature

If you experience **painful muscular spasms** (particularly in the legs, arms or abdomen, in many cases after sustained exercise during very hot weather),

- Rest immediately in a cool place, and drink oral rehydration solutions containing electrolytes
- Medical attention is needed if heat cramps last more than one hour

Heatstroke is a medical emergency!

Be aware of **Danger signs** & seek immediate medical attention if you observe

In adults

- Altered mental sensorium with disorientation, confusion and agitation, irritability, ataxia, seizure or coma
- Hot, red and dry skin
- Core body temperature $\geq 40^{\circ}\text{C}$ or 104°F
- Throbbing headache
- Anxiety, Dizziness, fainting and light headedness
- Muscle weakness or cramps
- Nausea and vomiting
- Rapid heartbeat/Rapid, shallow breathing

In children

- Refusal to feed
- Excessive irritability
- Decreased urine output
- Dry oral mucosa & absence of tear/sunken eyes
- Lethargy/altered sensorium
- Seizures
- Bleeding from any site

Call 108/102 immediately
if you find someone with
high body temperature and is either
unconscious, confused, or
has stopped sweating

Cool the person right away, while waiting, by:

- moving them to a cool place, if you can;
- applying cold water to large areas of the skin or clothing; and
- fanning the person as much as possible



National Programme
on Climate Change
and Human Health



जिल्हा आपत्ती व्यवस्थापन प्राधिकरण, नाशिक

उन्हाळ्यामध्ये उष्माघातापासून बचाव होण्याच्या दृष्टीने घ्यावयाची दक्षता

जिल्हा आपत्ती
व्यवस्थापन
प्राधिकरण

✓ काय करावे? (Do's)

1. भरपूर पाणी प्या: तहान लागलेली नसली तरीही, दर अर्ध्या तासाने पाणी प्यावे.
2. संरक्षणात्मक साधने: घराबाहेर पडताना छात्री, टोपी किंवा रुमालाचा वापर करा.
3. वेळेचे नियोजन: दुपारी १२ ते ३ वाजेदरम्यान शक्यतो घराबाहेरील कामे टाळावीत.
4. घराचे तापमान: खिडक्यांना पडदे लावा आणि दुपारी झडपा बंद ठेवा जेणेकरून घर थंड राहील.
5. हवामान अपडेट्स: रेडिओ, टीव्ही, वर्तमानपत्रे किंवा सोशल मीडियाद्वारे लक्ष ठेवा.
6. कपड्यांची निवड: हलकी, सुती आणि फिकट रंगाची कपडे वापरावीत.
7. प्रवासात काळजी: प्रवासात पिण्याचे पाणी नेहमी सोबत ठेवावे.
8. कष्टाचे काम: उन्हात काम करावे लागल्यास डोके, चेहरा आणि मान झाकून ठेवा.

9. पेयांचे सेवन: शरीरातील पाण्याचे प्रमाण टिकवणासाठी ORS, लस्सी, पन्हे, लिंबू-पाणी किंवा ताक नियमित घ्या.
10. लक्षणे ओळखा: चक्कर येणे, मळमळणे किंवा सतत डोकेदुखी झाल्यास तात्काळ डॉक्टरांकडे जा.
11. पशुधनाची काळजी: गुरांना सावलीत (छावणीत) ठेवावे आणि त्यांना पुरेसे पाणी द्यावे.
12. हवेशीर वातावरण: रात्रीच्या वेळी खिडक्या उघड्या ठेवा जेणेकरून हवा खेळती राहील.
13. थंडाव्यासाठी उपाय: पंखे, कुलर आणि ओल्या कपड्यांचा वापर करा.
14. कामाच्या ठिकाणी सोयी: कामगारांसाठी पिण्याचे पाणी आणि सावलीत विश्रांतीची सोय करा.
15. सूर्यप्रकाश टाळा: कामगारांना थेट सूर्यप्रकाश टाळण्यासाठी सूचित करा.
16. कामाची वेळ: शारीरिक कष्टाची कामे शक्यतो पहाटे किंवा सायंकाळी करावीत. अंधूनमथून विश्रांती घ्यावी.
17. विशेष काळजी: गरोदर स्त्रिया आणि आजारी व्यक्तींची अधिक काळजी घ्यावी.

✗ काय करू नये? (Don'ts)

1. उन्हात अतिकष्टाची कामे करू नका.
2. दारू, चहा, कॉफी आणि कार्बोनेटेड थंड पेये घेऊ नका.
3. दुपारी १२ ते ३ च्या दरम्यान घराबाहेर जाणे टाळा.
4. उच्च प्रथिनयुक्त आणि शिळे अन्न खाऊ नका.
5. लहान मुले किंवा पाळीव प्राण्यांना बंद आणि पार्क केलेल्या वाहनात सोडू नका.
6. गडद रंगाचे, घट्ट आणि कृत्रिम धाग्यांचे (Synthetic) कपडे घालणे टाळावे.
7. बाहेर तापमान जास्त असल्यास शारीरिक श्रमाची कामे टाळावी.
8. उन्हाच्या तीव्रतेच्या काळात स्वयंपाक करणे टाळावे.



जिल्हा आपत्ती व्यवस्थापन
प्राधिकरण, नाशिक
सतर्क रहा, सुरक्षित रहा!